Supplement to "Inference on Dynamic Spatial Autoregressive Models with Change Point Detection"

In this supplementary material, we provide additional mathematical details for Section 2.2 in the main text, extra simulation results, and additional explanations for the real data analysis. Proofs of the main results stated in the main text are also included, together with auxiliary results and their proofs.

S1. ADDITIONAL DETAILS FOR THE MODEL

S1.1. Additional details for Section 2.2

Together with (2.11), the least squares problem in (2.6) can be described as

$$\widetilde{\boldsymbol{\phi}} = \operatorname*{argmin}_{\boldsymbol{\phi}} \frac{1}{2T} \Big\| \mathbf{B}^\top \mathbf{y} - \mathbf{B}^\top \mathbf{V} \boldsymbol{\phi} - \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}(\boldsymbol{\phi})} \mathrm{vec}\big(\mathbf{I}_d\big) \Big\|^2. \tag{S1.1}$$

With the least squares estimator $\widetilde{\boldsymbol{\phi}}$, the problem in (2.7) in matrix notation is

$$\widehat{\boldsymbol{\phi}} = \operatorname*{argmin}_{\boldsymbol{\phi}} \frac{1}{2T} \Big\| \mathbf{B}^\top \mathbf{y} - \mathbf{B}^\top \mathbf{V} \boldsymbol{\phi} - \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}(\boldsymbol{\phi})} \mathrm{vec}\big(\mathbf{I}_d\big) \Big\|^2 + \lambda \mathbf{u}^\top |\boldsymbol{\phi}|, \tag{S1.2}$$
$$\text{subj. to } \|\boldsymbol{\Lambda}_t \boldsymbol{\Phi}\|_\infty < 1, \text{ with } |\mathbf{z}_t^\top \boldsymbol{\phi}| < 1 \text{ for any } t \in [T].$$

Note the squared errors in (S1.1) and (S1.2) are still implicit in $\boldsymbol{\phi}$ due to $\boldsymbol{\beta}(\boldsymbol{\phi})$.

Note from (2.13) and the definition of $\mathbf{B}$ in (2.9), we have

$$\begin{aligned}
&\mathbf{B}^\top \Big(\mathbf{I}_d \otimes \Big\{ \Big(\mathbf{I}_T \otimes \Big\{ (\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \Big\} \Big) (\mathbf{X}_1, \ldots, \mathbf{X}_T)^\top \Big\} \Big) \mathrm{vec}\big(\mathbf{I}_d\big) \\
&= T^{-1/2} d^{-a/2} \Big(\mathbf{I}_d \otimes \Big\{ (\mathbf{B}_1 - \bar{\mathbf{B}}, \ldots, \mathbf{B}_T - \bar{\mathbf{B}}) (\mathbf{I}_T \otimes \boldsymbol{\gamma}) \\
&\quad \Big(\mathbf{I}_T \otimes \Big\{ (\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \Big\} \Big) (\mathbf{X}_1, \ldots, \mathbf{X}_T)^\top \Big\} \Big) \mathrm{vec}\big(\mathbf{I}_d\big) \\
&= T^{-1/2} d^{-a/2} \Big(\mathbf{I}_d \otimes \Big\{ \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} (\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \mathbf{X}_t^\top \Big\} \Big) \mathrm{vec}\big(\mathbf{I}_d\big) \\
&= T^{-1/2} d^{-a/2} \mathrm{vec}\big(\sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} (\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \mathbf{X}_t^\top \big)
\end{aligned}$$

$$
\begin{aligned}
&= T^{-1/2} d^{-a/2} \Big(\sum_{t=1}^{T} \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \Big) \mathrm{vec}\Big((\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \Big) \\
&= T^{-1/2} d^{-a/2} \Big(\sum_{t=1}^{T} \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \Big) \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{y}^\nu .
\end{aligned}
$$

Similarly, by the definition of $\mathbf{Y}_W$,

$$
\begin{aligned}
&\mathbf{B}^\top \Big(\mathbf{I}_d \otimes \Big\{ \Big(\mathbf{I}_T \otimes \Big\{ \Big(\sum_{j=1}^{p} \sum_{k=0}^{l_j} \phi_{j,k} (\mathbf{y}^\nu_{j,k})^\top (\mathbf{W}^\otimes_j)^\top \Big) \\
&\qquad \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \Big\} \Big) (\mathbf{X}_1, \ldots, \mathbf{X}_T)^\top \Big\} \Big) \mathrm{vec}\big(\mathbf{I}_d \big) \\
&= T^{-1/2} d^{-a/2} \Big(\mathbf{I}_d \otimes \Big\{ (\mathbf{B}_1 - \bar{\mathbf{B}}, \ldots, \mathbf{B}_T - \bar{\mathbf{B}}) (\mathbf{I}_T \otimes \boldsymbol{\gamma}) \\
&\quad \cdot \Big(\mathbf{I}_T \otimes \Big\{ \Big(\sum_{j=1}^{p} \sum_{k=0}^{l_j} \phi_{j,k} (\mathbf{y}^\nu_{j,k})^\top (\mathbf{W}^\otimes_j)^\top \Big) \\
&\quad \cdot \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \Big\} \Big) (\mathbf{X}_1, \ldots, \mathbf{X}_T)^\top \Big\} \Big) \mathrm{vec}\big(\mathbf{I}_d \big) \\
&= T^{-1/2} d^{-a/2} \Big(\mathbf{I}_d \otimes \Big\{ \sum_{t=1}^{T} (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \Big(\sum_{j=1}^{p} \sum_{k=0}^{l_j} \phi_{j,k} (\mathbf{y}^\nu_{j,k})^\top (\mathbf{W}^\otimes_j)^\top \Big) \\
&\quad \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \mathbf{X}_t^\top \Big\} \Big) \mathrm{vec}\big(\mathbf{I}_d \big) \\
&= T^{-1/2} d^{-a/2} \mathrm{vec}\Big(\sum_{t=1}^{T} (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} (\boldsymbol{\phi}^\top \mathbf{Y}_W^\top) \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \mathbf{X}_t^\top \Big) \\
&= T^{-1/2} d^{-a/2} \Big(\sum_{t=1}^{T} \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \Big) \big[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \big]^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{Y}_W \boldsymbol{\phi} .
\end{aligned}
$$

With (2.13), the term inside the squared Euclidean norm of (S1.1) can hence be further written as

$$
\begin{aligned}
&\mathbf{B}^\top\mathbf{y}-\mathbf{B}^\top\mathbf{V}\boldsymbol{\phi}-\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}(\boldsymbol{\phi})}\mathrm{vec}\big(\mathbf{I}_d\big)\\
=&\mathbf{B}^\top\mathbf{y}-\mathbf{B}^\top\mathbf{V}\boldsymbol{\phi}-\mathbf{B}^\top\Big(\mathbf{I}_d\otimes\Big\{\Big(\mathbf{I}_T\otimes\Big\{(\mathbf{y}^\nu)^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\big[\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\big]^{-1}\Big\}\Big)\\
&\cdot(\mathbf{X}_1,\dots,\mathbf{X}_T)^\top\Big\}\Big)\mathrm{vec}\big(\mathbf{I}_d\big)\\
&+\mathbf{B}^\top\Big(\mathbf{I}_d\otimes\Big\{\Big(\mathbf{I}_T\otimes\Big\{\Big(\sum_{j=1}^{p}\sum_{k=0}^{l_j}\phi_{j,k}(\mathbf{y}^\nu_{j,k})^\top(\mathbf{W}^\otimes_j)^\top\Big)\\
&\quad\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\big[\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\big]^{-1}\Big\}\Big)(\mathbf{X}_1,\dots,\mathbf{X}_T)^\top\Big\}\Big)\mathrm{vec}\big(\mathbf{I}_d\big)\\
=&\mathbf{B}^\top\mathbf{y}-T^{-1/2}d^{-a/2}\Big(\sum_{t=1}^{T}\mathbf{X}_t\otimes(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\Big)\big[\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\big]^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{y}^\nu\\
&-\Big\{\mathbf{B}^\top\mathbf{V}-T^{-1/2}d^{-a/2}\Big(\sum_{t=1}^{T}\mathbf{X}_t\otimes(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\Big)\big[\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\big]^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{Y}_W\Big\}\boldsymbol{\phi}\\
=&\mathbf{B}^\top\mathbf{y}-\boldsymbol{\Xi}\mathbf{y}^\nu-(\mathbf{B}^\top\mathbf{V}-\boldsymbol{\Xi}\mathbf{Y}_W)\boldsymbol{\phi}.
\end{aligned}
\tag{S1.3}
$$

S1.2. Identification of the model (2.11)

We show that the coefficients $\boldsymbol{\phi}^*$ and $\boldsymbol{\beta}^*$ in model (2.11) are identified under Assumption (I1). Assume that the two sets of parameters $(\check{\boldsymbol{\phi}},\check{\boldsymbol{\beta}})$ and $(\acute{\boldsymbol{\phi}},\acute{\boldsymbol{\beta}})$ both satisfy model (2.11), then we have

$$\mathbf{B}^\top\mathbf{V}\check{\boldsymbol{\phi}}+\mathbf{B}^\top\mathbf{X}_{\check{\boldsymbol{\beta}}}\mathrm{vec}\big(\mathbf{I}_d\big)=\mathbf{B}^\top\mathbf{V}\acute{\boldsymbol{\phi}}+\mathbf{B}^\top\mathbf{X}_{\acute{\boldsymbol{\beta}}}\mathrm{vec}\big(\mathbf{I}_d\big).$$

By noticing that $\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}}\mathrm{vec}\big(\mathbf{I}_d\big)=\mathbf{B}^\top\widetilde{\mathbf{X}}\boldsymbol{\beta}$, we may rearrange the above and arrive at

$$\mathbf{0}=\mathbf{B}^\top\mathbf{V}(\check{\boldsymbol{\phi}}-\acute{\boldsymbol{\phi}})+\mathbf{B}^\top\widetilde{\mathbf{X}}(\check{\boldsymbol{\beta}}-\acute{\boldsymbol{\beta}})=\begin{pmatrix}\mathbf{B}^\top\mathbf{V} & \mathbf{B}^\top\widetilde{\mathbf{X}}\end{pmatrix}\begin{pmatrix}\check{\boldsymbol{\phi}}-\acute{\boldsymbol{\phi}}\\ \check{\boldsymbol{\beta}}-\acute{\boldsymbol{\beta}}\end{pmatrix}.$$

Taking expectation and left-multiplying by $(\mathbf{Q}^\top\mathbf{Q})^{-1}\mathbf{Q}^\top$ on both sides, we obtain $\check{\boldsymbol{\phi}}=\acute{\boldsymbol{\phi}}$ and $\check{\boldsymbol{\beta}}=\acute{\boldsymbol{\beta}}$. This implies $\boldsymbol{\phi}^*$ and $\boldsymbol{\beta}^*$ are identified in model (2.1), and hence $\boldsymbol{\mu}^*$ is uniquely identified.

It is worth to remind that in practice, we should be careful in constructing dynamic variables to due to the issue of multicollinearity, as reflected in $\mathbf{B}^\top\mathbf{V}$ above.

S1.3. Additional details to Section 5.1

In this subsection, we discuss the model framework with regime changes based on multiple threshold variables with multiple threshold values. For illustrations, consider the following spatial autoregressive model with $(k+1)$ regimes, where k can be unknown:

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \Big(\phi_1^* \mathbb{1}\{q_t \leq \gamma_1^*\} + \phi_2^* \mathbb{1}\{\gamma_1^* < q_t \leq \gamma_2^*\} + \cdots + \phi_k^* \mathbb{1}\{q_t > \gamma_k^*\}\Big) \mathbf{W}_1 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t. \tag{S1.4}$$

Model (S1.4) can be written in the form of (5.5), with the consistency of $\{\phi_{1,l}^*\}_{l\in[L]}$ guaranteed by Corollary S1. This also implies that the estimation on k is consistent.

Corollary S1 *(Consistency on the number of threshold regimes and multiple threshold values estimation) Let all assumptions in Theorem 2 hold. For model (S1.4), let $\{\widehat{\phi}_{1,l}\}_{l\in[L]}$ denote the adaptive LASSO solution for $\{\phi_{1,l}^*\}_{l\in[L]}$ in (5.5). Then, $\widehat{k} := |\{\gamma_l : \widehat{\phi}_{1,l} \neq 0\}|$ estimates k consistently. Moreover, for every $i \in [\widehat{k}]$, the i-th smallest element in $\{\gamma_l : \widehat{\phi}_{1,l} \neq 0\}$ estimates γ_i^* consistently.*

As we often have limited prior knowledge on the parameter space in practice, we recommend using our framework in an exploratory way. This should help researchers discover more reasonable threshold structures in the data.

S2. ADDITIONAL SIMULATIONS

S2.1. Different methods to select the tuning parameter

In this subsection, we conduct Monte Carlo experiments to study the sensitivity of the estimator's performance with respect to different methods for selecting the tuning parameter λ in the adaptive LASSO problem. Specifically, we compare our BIC criterion in (4.1) with two alternative choices: (i) three-fold cross validation minimizing the least squares error, which effectively uses the first term in (4.1) as the criterion (five-fold and ten-fold cross validation are also experimented, yielding similar results and thus omitted here); (ii) the BIC criterion inspired by Wang et al. (2009) (cf. equation (2.7) within):

$$\mathrm{BIC}(\lambda) = \log\Big(\frac{1}{T}\Big\|\mathbf{B}^\top\mathbf{y} - \mathbf{B}^\top\mathbf{V}\widehat{\boldsymbol{\phi}} - \mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}(\widehat{\boldsymbol{\phi}})}\mathrm{vec}\big(\mathbf{I}_d\big)\Big\|^2\Big) + |\widehat{H}|\frac{\log(T)}{T}\log(\log(L)), \tag{S2.1}$$

with variables constructed under the model framework (2.1), which an alternative choice to account for potentially growing L.

We use a similar data generating process to (6.1):

$$\mathbf{y}_t = \Big\{\mathbf{I}_d - \big(0+0.2z_{1,1,t}+0z_{1,2,t}\big)\mathbf{W}_1 - \big(0+0.2z_{2,1,t}+0z_{2,2,t}\big)\mathbf{W}_2\Big\}^{-1}\big(\boldsymbol{\mu}^* + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t\big), \qquad \text{(S2.2)}$$

where $\{z_{1,1,t}\}$, $\{z_{1,2,t}\}$, $\{z_{2,1,t}\}$, $\{z_{2,2,t}\}$, $\boldsymbol{\mu}^*$, $\mathbf{X}_t$, $\boldsymbol{\beta}^*$, $\mathbf{W}_1$, $\mathbf{W}_2$, and $\boldsymbol{\epsilon}_t$ are generated in the same way as described after (6.1) in the main text, and $z_{1,0,t} = z_{2,0,t} = 1$ for all $t \in [T]$. We first fit the model using these dynamic variables, i.e., $L=6$. For a more thorough comparison between the two criteria in (4.1) and (S2.1), we also perform estimation using additional dynamic variables $\{z_{1,3,t}\},\ldots,\{z_{1,9,t}\},\{z_{2,3,t}\},\ldots,\{z_{2,9,t}\}$, all generated independently from $\mathcal{N}(0,1)$. In this case, we have that $L=20$ and that the true parameters corresponding to these extra dynamic variables are all zero. For both scenarios, we experiment with $T=50$ and $d=25,50$, respectively, and each setting is repeated 500 times. Table S1 reports the error measures of the parameter estimators for all settings under $L=6$, and Table S2 for $L=20$.

	$d=25$			$d=50$		
	CV	BIC	BIC2	CV	BIC	BIC2
$\widehat{\boldsymbol{\phi}}$ MSE	.001	.000	.000	.001	.000	.000
	(.001)	(.001)	(.000)	(.001)	(.000)	(.000)
$\widehat{\boldsymbol{\beta}}$ MSE	.001	.001	.001	.001	.001	.001
	(.000)	(.000)	(.000)	(.000)	(.000)	(.000)
$\widehat{\boldsymbol{\mu}}$ MSE	.021	.021	.021	.022	.021	.021
	(.005)	(.004)	(.004)	(.005)	(.004)	(.004)
$\widehat{\boldsymbol{\phi}}$ Specificity	.463	.743	.743	.467	.745	.745
	(.326)	(.326)	(.326)	(.328)	(.325)	(.325)
$\widehat{\boldsymbol{\phi}}$ Sensitivity	1.000	.999	1.000	1.000	1.000	1.000
	(.000)	(.022)	(.000)	(.000)	(.000)	(.000)

Table S1: Simulation results for different methods of selecting the tuning parameter. The data generating process follows (S2.2) with $T=50$, and estimation is performed using $L=6$ dynamic variables. The columns labeled "CV", "BIC", and "BIC2" correspond to selection based on least squares residuals, (4.1), and (S2.1), respectively. The mean and standard deviation (in brackets) of the corresponding error measures over 500 repetitions are reported.

Both tables show that selecting λ using either of the two BIC criteria greatly outperforms selection based on the least squares criterion ("CV") in terms of specificity, while all three methods achieve high sensitivity. This is reasonable, as the BIC criteria are specifically designed to address

	$d=25$			$d=50$		
	CV	BIC	BIC2	CV	BIC	BIC2
$\widehat{\boldsymbol{\phi}}$ MSE	.002	.001	.001	.006	.001	.001
	(.001)	(.001)	(.001)	(.004)	(.002)	(.002)
$\widehat{\boldsymbol{\beta}}$ MSE	.001	.001	.001	.001	.001	.001
	(.001)	(.001)	(.001)	(.000)	(.000)	(.000)
$\widehat{\boldsymbol{\mu}}$ MSE	.023	.021	.021	.021	.020	.020
	(.007)	(.006)	(.006)	(.005)	(.004)	(.004)
$\widehat{\boldsymbol{\phi}}$ Specificity	.489	.886	.886	.495	.906	.906
	(.196)	(.142)	(.142)	(.196)	(.114)	(.114)
$\widehat{\boldsymbol{\phi}}$ Sensitivity	.996	.959	.960	1.000	1.000	1.000
	(.045)	(.137)	(.136)	(.000)	(.000)	(.000)

Table S2: Simulation results for the same data generating process as in Table S1, with model fitting performed using $L=20$ dynamic variables.

model overfitting and are thus less likely to select truly zero parameters. We also observe that the results under the two BIC criteria are very similar, which is expected since even $L=20$ in Table S2 is still considered small. Both tables demonstrate the strength of our approach and suggest that in practice, "BIC" may be used effectively.

S2.2. Sensitivity of parameter estimators to endogeneity

In the main text, we demonstrated the effectiveness of our parameter estimators under mild endogeneity. To further investigate the impact of endogeneity, we consider the model

$$\mathbf{y}_t=\Big\{\mathbf{I}_d-\big(0+0.2z_{1,1,t}+0z_{1,2,t}\big)\mathbf{W}_1-\big(0+0.2z_{2,1,t}+0z_{2,2,t}\big)\mathbf{W}_2\Big\}^{-1}\big(\boldsymbol{\mu}^*+\mathbf{X}_t\boldsymbol{\beta}^*+\boldsymbol{\epsilon}_t\big), \qquad \text{(S2.3)}$$

where $\{z_{1,1,t}\}$, $\{z_{1,2,t}\}$, $\{z_{2,1,t}\}$, $\{z_{2,2,t}\}$, $\boldsymbol{\mu}^*$, $\boldsymbol{\beta}^*$, $\mathbf{W}_1$, $\mathbf{W}_2$, and $\boldsymbol{\epsilon}_t$ are again generated in the same way as described after (6.1) in the main text. The covariate $\mathbf{X}_t$ also has three features, with each entry independently drawn from the standard normal distribution, except that the third feature is made endogenous by adding $m\,\boldsymbol{\epsilon}_t$. Thus, the parameter m controls the level of endogeneity, and we experiment with values of m ranging from 0 to 100. For computational convenience, we set the grid of the tuning parameter to be a neighborhood around the value of λ that minimizes the least squares criterion "CV" in Section S2.1. The results for the dimension settings $(T,d)=(50,25)$ and $(50,50)$ are summarized in Figure S1 and Figure S2, respectively.

The performance of all estimators improves slightly as the spatial dimension increases, although

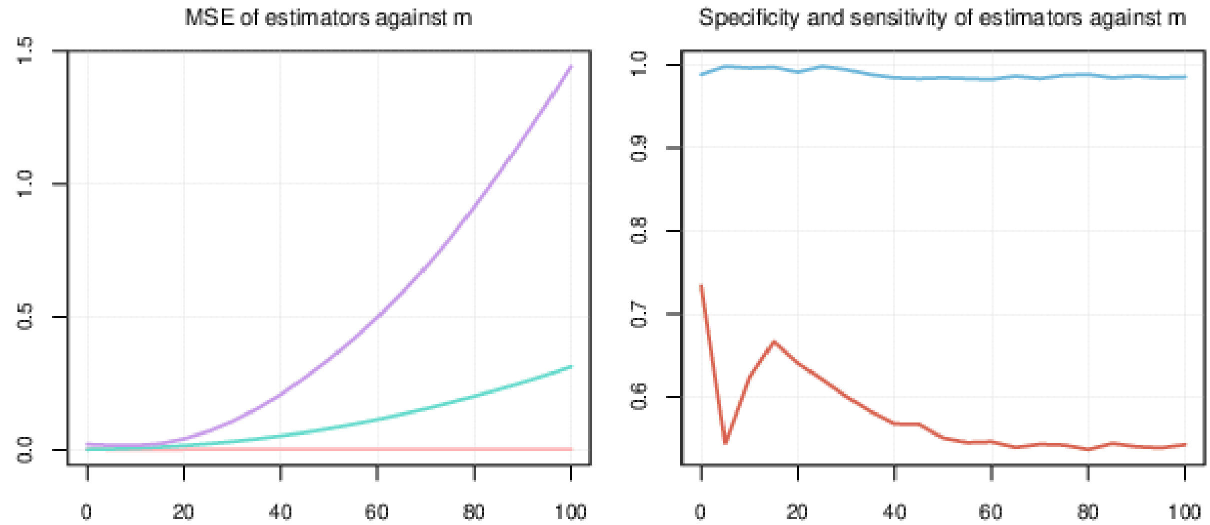

Figure S1: Estimator performance for model (S2.3) with m varying from 0 to 100, under the setting $(T,d)=(50,25)$. The left panel shows the MSE for $\widehat{\boldsymbol{\phi}}$ (pink), $\widehat{\boldsymbol{\beta}}$ (teal), and $\widehat{\boldsymbol{\mu}}$ (purple); the right panel shows the specificity (red) and sensitivity (blue) of $\widehat{\boldsymbol{\phi}}$. Each value is averaged over 500 repetitions.

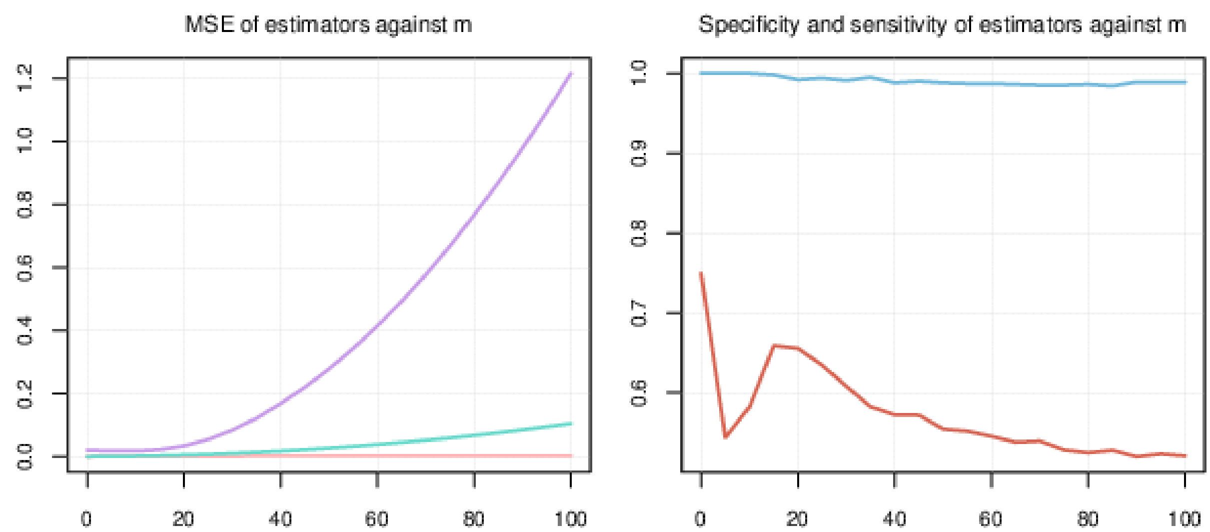

Figure S2: Estimator performance for model (S2.3) with m varying from 0 to 100, under the setting $(T,d)=(50,50)$.

the general patterns remain consistent across both settings. From the left panels in both figures, we observe that all MSE's worsen as the covariates become more endogenous. Notably, the performance of the estimator $\widehat{\boldsymbol{\mu}}$ is most affected by endogeneity, as the estimation errors in $\widehat{\boldsymbol{\phi}}$ and $\widehat{\boldsymbol{\beta}}$ propagate into $\widehat{\boldsymbol{\mu}}$. The right panels of both figures show that while the sensitivity of $\widehat{\boldsymbol{\phi}}$ remains consistently high, its specificity gradually declines as endogeneity increases, as one would expect. However, we note that the specificity curve exhibits a marked dip when m is relatively small. Each point on the two curves is computed using the value of λ that minimizes the BIC within a grid constructed based on the least squares criterion ("CV" in Section S2.1). It turns out that this grid can contain relatively large values and may not always provide a good approximation to the optimal λ. When a more carefully designed grid, typically with a wider range, is used, this dip can be largely mitigated, as illustrated in Figure S3 below. As with all penalization-based methods, this highlights both the importance and the challenge of specifying an appropriate grid of candidate tuning parameters.

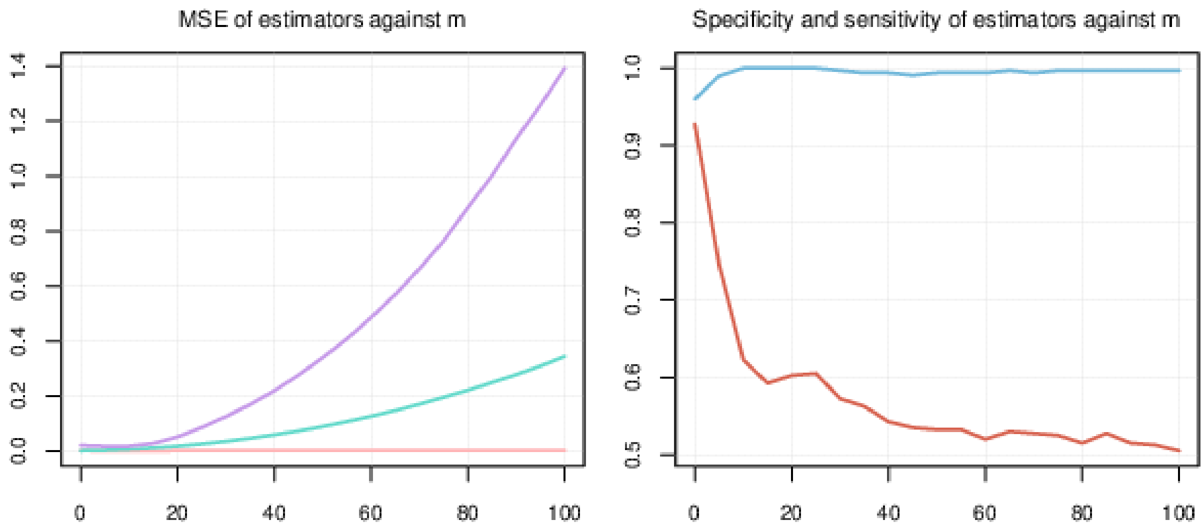

Figure S3: Similar to Figure S3, with a wider-range grid for the tuning parameter and each value is averaged over 100 repetitions.

S2.3. Experiments on the threshold spatial autoregressive models

We present numerical results for the threshold spatial autoregressive model introduced in Section 5.1. Specifically, we consider the data generating process

$$\mathbf{y}_t = \left(\mathbf{I}_d - 0.3 z_{1,t}\mathbf{W}_1 - 0.8 z_{2,t}\mathbf{W}_2\right)^{-1}\left(\boldsymbol{\mu}^* + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t\right), \tag{S2.4}$$

where $\mathbf{W}_1$, $\mathbf{W}_2$, $\boldsymbol{\mu}^*$, $\mathbf{X}_t$, $\boldsymbol{\beta}^*$ and $\boldsymbol{\epsilon}_t$ are generated in the same way as in (6.1). The indicators $z_{1,t} = \mathbb{1}\{q_t \leq \gamma^*\}$ and $z_{2,t} = \mathbb{1}\{q_t > \gamma^*\}$ are constructed using a threshold variable q_t and threshold value γ^*. Thus, (S2.4) defines a threshold spatial autoregressive model that has changes in both the coefficients and the spatial weight matrices. To simultaneously estimate the parameters and the threshold value, we fit a model of the form

$$\mathbf{y}_t=\Big(\mathbf{I}_d-\sum_{l=1}^{19}z_{1,l,t}\phi_{1,l}^*\mathbf{W}_1-\sum_{l=1}^{19}z_{2,l,t}\phi_{2,l}^*\mathbf{W}_2\Big)^{-1}(\boldsymbol{\mu}^*+\mathbf{X}_t\boldsymbol{\beta}^*+\boldsymbol{\epsilon}_t),\text{ where} \tag{S2.5}$$
$$z_{1,l,t}=\mathbb{1}\{q_t\le\widehat{\gamma}_l\},\ z_{2,l,t}=\mathbb{1}\{q_t>\widehat{\gamma}_l\},\ \widehat{\gamma}_l=(5\%\cdot l)\text{-th empirical quantile of }\{q_t\}_{t\in[T]}.$$

As discussed in Section 5.1, we expect $\phi_{1,1}^*,\ldots,\phi_{1,19}^*$ to be all zero, except for the one corresponding to $z_{1,l,t}$ where $\widehat{\gamma}_l$ is closest to the true threshold γ^*. Let l^* denote the index of this $\widehat{\gamma}_l$. Similarly, among $\phi_{2,1}^*,\ldots,\phi_{2,19}^*$, all entries should be zero except for the one corresponding to $z_{2,l^*,t}$. Moreover, we expect $(\phi_{1,l^*}^*,\phi_{2,l^*}^*)\approx(0.3,0.8)$.

We consider two types of threshold variables in our experiments:

1. *(AR(5))* q_t follows an AR(5) process with i.i.d. $\mathcal{N}(0,1)$ innovations, with $\gamma^*=0.3$;
2. *(Self-exciting on mean)* $q_t=\mathbf{1}^\top\mathbf{y}_{t-1}/d$, i.e., the regime switches are driven in a self-exciting manner based on the mean of the previous observation, with $\gamma^*=1.5$.

Results for $d=50,75$ and $T=100,150$ are reported in Table S3, where the estimators $\widehat{\boldsymbol{\phi}}$ and $\widehat{\gamma}_l$ are obtained from fitting the model (S2.5). The evaluation metrics are defined as follows:

$$\widehat{\boldsymbol{\phi}}\text{ MSE}:=\text{Mean squared error of }\widehat{\boldsymbol{\phi}}\text{ with }\boldsymbol{\phi}\text{ all zero except for }(\phi_{1,l^*}^*,\phi_{2,l^*}^*)=(0.3,0.8),$$
$$\widehat{\boldsymbol{\phi}}\text{ True-Unique}:=\mathbb{1}\{\widehat{\phi}_{1,l^*}\text{ and }\widehat{\phi}_{2,l^*}\text{ are uniquely identified as the only nonzero in }\widehat{\boldsymbol{\phi}}\},$$
$$\widehat{\gamma}_l\text{ MSE}:=\text{Mean squared error of }\widehat{\gamma}_l\text{ relative to the true threshold value }\gamma^*.$$

The metric $\widehat{\boldsymbol{\phi}}$ True-Unique is crucial for demonstrating the validity of our algorithm, as it reflects both specificity and sensitivity. In addition, it assesses whether the estimated threshold value is unique. When computing the MSE of $\widehat{\gamma}_l$ in cases where multiple indices l are identified with different $\widehat{\gamma}_l$ values, we select the index l corresponding to the largest $\widehat{\phi}_{1,l^*}$.

Table S3 confirms that our procedure can efficiently estimate both the threshold value simultaneously in one go. Although the estimator of γ^* is coarse, accurate only up to the 5% empirical quantile of the threshold variable, the results show that increasing the data dimensions improves the performance of $\widehat{\gamma}_l$. In practice, re-estimation using a finer grid based on such initial threshold estimate could further enhance accuracy.

q_t setting	AR(5)				Self-exciting on mean			
(T,d)	(100,50)	(100,75)	(150,50)	(150,75)	(100,50)	(100,75)	(150,50)	(150,75)
$\widehat{\boldsymbol{\phi}}$ MSE	.002	.003	.017	.020	.009	.007	.003	.001
	(.002)	(.003)	(.013)	(.011)	(.011)	(.010)	(.006)	(.003)
$\widehat{\boldsymbol{\phi}}$ True-Unique	.562	.439	.706	.844	.537	.621	.797	.938
	(.497)	(.500)	(.456)	(.363)	(.499)	(.485)	(.403)	(.242)
$\widehat{\gamma}_l$ MSE	.343	.066	.207	.025	.128	.031	.080	.005
	(.870)	(.309)	(.707)	(.193)	(.881)	(.126)	(.802)	(.023)

Table S3: Simulation results for fitting the threshold model (S2.5). The mean and standard deviation (in brackets) of the corresponding error measures over 500 repetitions are reported.

Comparison with Li and Lin (2024)

We also compare our method with the quasi-maximum likelihood estimator (QMLE) for the threshold spatial autoregressive model (TSAR) in panel data, which is a variant of the TSAR model in Li and Lin (2024) adapted to the time series setup. We consider the model

$$\mathbf{y}_t = \begin{cases} \boldsymbol{\mu}^* + (\rho^* + \varrho^*)\mathbf{W}_1\mathbf{y}_t + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & q_t \le \gamma^*; \\ \boldsymbol{\mu}^* + \rho^*\mathbf{W}_1\mathbf{y}_t + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & q_t > \gamma^*. \end{cases} \tag{S2.6}$$

This model represents a change in regime for the spatial correlation coefficients, from $(\rho^* + \varrho^*)$ to ρ^*, controlled by the threshold variable $\{q_t\}$. This setup mirrors the one for change point models in Section 6.2 of the main paper. Specifically, we generate $\boldsymbol{\mu}^*$, $\mathbf{W}_1$, $\boldsymbol{\mu}^*$, $\mathbf{X}_t$, $\boldsymbol{\beta}^*$ and $\boldsymbol{\epsilon}_t$ as described previously in (S2.4), and also generate q_t using both an AR(5) process and in a self-exciting manner, respectively. We fix $T = 100$, $\gamma^* = 0.05$, and $(\rho^*, \varrho^*) = (-0.3, 0.3)$, and experiment with three different methods, as described below:

1. AL: The adaptive LASSO estimator with dynamic variables $\{z_{t,0,1},\ldots,z_{t,9,1}\}$, where $z_{t,0,1} = 1$, $z_{t,k,1} = \mathbb{1}\{q_t \le \widehat{\gamma}_k\}$ for $k \in [9]$, $t \in [100]$, and $\widehat{\gamma}_k$ is the $(10\% \cdot k)$-th empirical quantile of $\{q_t\}_{t\in[100]}$.
2. AL-DC: The adaptive LASSO estimator using the divide-and-conquer scheme in Remark 1, with three subsets $\{z_{t,0,1}, z_{t,1,1}, z_{t,2,1}, z_{t,3,1}\}$, $\{z_{t,0,1}, z_{t,4,1}, z_{t,5,1}, z_{t,6,1}\}$, and $\{z_{t,0,1}, z_{t,7,1}, z_{t,8,1}, z_{t,9,1}\}$, where all dynamic variables are the same as in method 1 above. The final estimator is selected from the three results with the smallest estimation error.
3. QMLE (Li and Lin, 2024): The estimation is performed for each $\{\widehat{\gamma}_k\}_{k\in[9]}$, where $\widehat{\gamma}_k$ is the same as in AL. The optimization step is carried out using the Nelder–Mead algorithm, and the final estimator is selected as the one with the smallest estimation error among them.

Table S4 reports the results for $T = 100$ and $d = 25, 50$, where all simulations are repeated 500 times.

$\boldsymbol{\epsilon}_t$	AR(5)			Self-exciting on mean		
$d=25$	AL	AL-DC	QMLE	AL	AL-DC	QMLE
$\widehat{\boldsymbol{\phi}}$ MSE	.036	**.011**	.027	.037	**.014**	.030
	(.041)	(.015)	(.011)	(.038)	(.016)	(.089)
Estimation Error	1.00	**.994**	1.05	1.00	**.994**	1.21
	(.031)	(.028)	(.040)	(.031)	(.028)	(.492)
Run Time (s)	**1.57**	2.00	15.1	**1.40**	1.78	12.2
	(.271)	(.330)	(3.85)	(.088)	(.080)	(.240)
$d=50$	AL	AL-DC	QMLE	AL	AL-DC	QMLE
$\widehat{\boldsymbol{\phi}}$ MSE	.012	**.004**	.030	.013	**.006**	.321
	(.011)	(.006)	(.011)	(.011)	(.008)	(.279)
Estimation Error	.988	**.986**	1.04	.992	**.991**	2.91
	(.020)	(.020)	(.031)	(.019)	(.019)	(1.58)
Run Time (s)	**6.60**	8.25	44.3	**6.52**	8.16	44.1
	(.109)	(.090)	(.681)	(.103)	(.095)	(.861)

Table S4: Simulation results for the model (S2.6) with $T=100$ using three methods: AL, AL-DC, and QMLE. The mean and standard deviation (in brackets) of the measures over 500 repetitions are reported.

From this table, we observe that our adaptive LASSO estimator with the divide-and-conquer scheme achieves the best estimation accuracy while having high computational efficiency. We thus reach the same conclusions as in Section 6.2 for Tables 3 and 4.

S2.4. Experiments on the divide-and-conquer scheme in Remark 1

We demonstrate here the numerical performance of the divide-and-conquer scheme depicted in Remark 1. Under $(T,d)=(100,75)$, consider an extension of (6.2) with two change points:

$$\mathbf{y}_t = \big(\mathbf{I}_d - 0.8\mathbb{1}_{\{t\le 30\}}\mathbf{W}_1 + 0.9\mathbb{1}_{\{t\le 60\}}\mathbf{W}_1 + 0.9\mathbb{1}_{\{t>60\}}\mathbf{W}_2\big)^{-1}\big(\boldsymbol{\mu}^* + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t\big). \tag{S2.7}$$

That is, the spatial weight matrix changes from $-0.1\mathbf{W}_1$ to $-0.9\mathbf{W}_1$ at $t=30$, followed by a change from $-0.9\mathbf{W}_1$ to $-0.9\mathbf{W}_2$ at $t=60$. Suppose it is only known a priori that on $\mathcal{T}=\{2,4,\ldots,98,100\}$[1], the spatial weight matrix might change from $\mathbf{W}_1$ to $\mathbf{W}_2$ and the spatial correlation coefficients might change as well. We wish to estimate the number of changes and the change locations. Following Remark 1, we construct the ordered sets $\mathcal{T}_1=\{2,4,\ldots,20\}$, $\mathcal{T}_2=\{20,22,\ldots,40\}$, $\mathcal{T}_3=\{40,42,\ldots,60\}$, $\mathcal{T}_4=\{60,62,\ldots,80\}$ and $\mathcal{T}_5=\{80,82,\ldots,100\}$. Note that we add "overlaps" between adjacent sets

[1] Change point identified at the last observed time point $T=100$ represents no change in the structure.

to circumvent falsely identifying change candidates at the margin. Now, the size of each set is 11, which is reasonable according to the numerical results in Tables 2. For each $j \in [5]$, consider[2]

$$\mathbf{y}_t = \Big(\mathbf{I}_d - \sum_{l=1}^{|\mathcal{T}_j|} z_{j,1,l,t}\phi^*_{j,1,l}\mathbf{W}_1 - \sum_{l=1}^{|\mathcal{T}_j|} z_{j,2,l,t}\phi^*_{j,2,l}\mathbf{W}_2\Big)^{-1}\big(\boldsymbol{\mu}^* + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t\big), \text{ where} \tag{S2.8}$$

$$z_{j,1,l,t} := \mathbb{1}_{\{t \le (\mathcal{T}_j)_l\}},\ z_{j,2,l,t} := \mathbb{1}_{\{t > (\mathcal{T}_j)_l\}}.$$

As in Remark 1, all time points corresponding to nonzero estimates of $\phi^*_{j,1,l}$ or $\phi^*_{j,2,l}$ for $j \in [5], l \in [|\mathcal{T}_j|]$ are identified and collected to form a refined candidate set $\widetilde{\mathcal{T}}$, where marginal time points (20, 40, 60,80) are discarded if they are not identified in all $\mathcal{T}_j$ containing them. Finally, consider

$$\mathbf{y}_t = \Big(\mathbf{I}_d - \sum_{l=1}^{|\widetilde{\mathcal{T}}|} z_{1,l,t}\phi^*_{1,l}\mathbf{W}_1 - \sum_{l=1}^{|\widetilde{\mathcal{T}}|} z_{2,l,t}\phi^*_{2,l}\mathbf{W}_2\Big)^{-1}\big(\boldsymbol{\mu}^* + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t\big), \text{ where} \tag{S2.9}$$

$$z_{1,l,t} = \mathbb{1}_{\{t \le \widetilde{\mathcal{T}}_l\}},\ z_{2,l,t} = \mathbb{1}_{\{t > \widetilde{\mathcal{T}}_l\}}.$$

Then, change points are estimated as the timestamps corresponding to nonzero estimates for $\phi^*_{1,l}$ or $\phi^*_{2,l}$. The histogram for the estimated change locations over 500 repetitions is shown in the left panel of Figure S4 and is encouraging. To further quantify the performance of our scheme, we use the Adjusted Rand index (ARI) of the estimated time segmentation against the truth[3](Rand, 1971; Hubert and Arabie, 1985), a measure frequently used by change point researchers (Wang and Samworth, 2017). The average ARI across all runs is 0.901, again suggesting that our scheme is performing very well.

We also consider (S2.7) under no change or, equivalently, one change at $t = 100$:

$$\mathbf{y}_t = \big(\mathbf{I}_d + 0.9 \cdot \mathbb{1}_{\{t \le T\}}\mathbf{W}_1\big)^{-1}\big(\boldsymbol{\mu}^* + \mathbf{X}_t\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t\big). \tag{S2.10}$$

We follow the same exact procedure to estimate (S2.7), and the histogram for the estimated change points over 500 runs is shown in the right panel of Figure S4. In 98% of the experiments, exactly $T = 100$ is identified, meaning no change is detected, which corresponds to a 2% false change discovery rate. Furthermore, the average ARI[4] is 0.980.

[2]Note that the last dynamic variable in $\mathcal{T}_5$ for $\mathbf{W}_2$, $z_{5,2,11,t}$, is 0 for all t, so we directly specify $\phi^*_{5,2,11}$ as 0.

[3]The estimated time segmentation assigns the same labels to time points between the estimated change points, with different labels assigned after each change point. For true time partitioning with changes at {30,60}, the intervals {1,2,...,30}, {31,32,...,60}, and {61,62,...,100} are labelled as 1, 2, and 3, respectively.

[4]The true time partition, when there are no changes, labels all time points as 1.

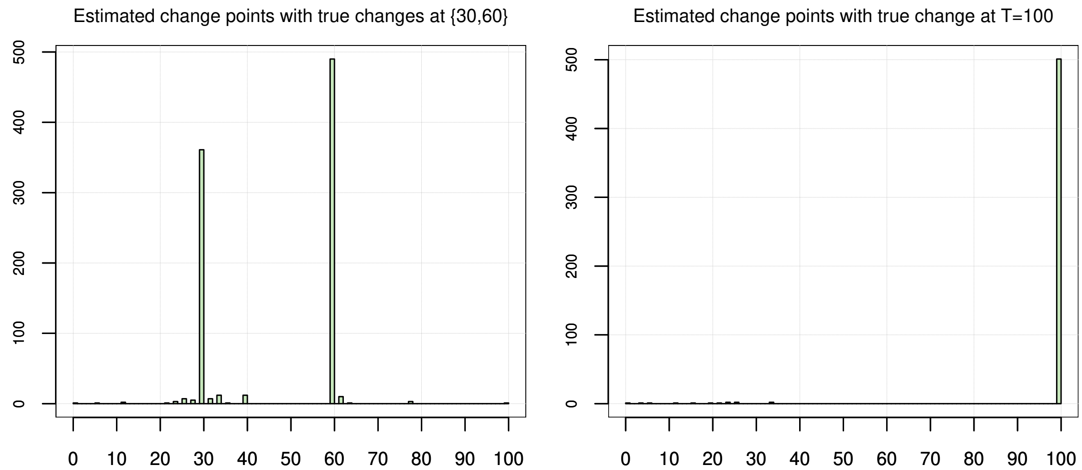

Figure S4: Histograms of estimated change locations under true models (S2.7) (left panel) and (S2.10) (right panel). Both experiments are repeated 500 times.

S3. ADDITIONAL REAL DATA EXAMPLE: ENTERPRISE PROFITS

In this case study, we use our proposed model to analyze the total profits of enterprises for a selection of provincial regions in China. Our panel data covers $T = 86$ monthly periods from March 2016 to August 2024 and 25 provinces and 4 direct-administered municipalities (i.e., $d = 29$).

Due to missingness, we exclude January and February data. The 25 provinces included in the data are Hebei, Shanxi, Inner Mongolia, Liaoning, Jilin, Heilongjiang, Jiangsu, Zhejiang, Anhui, Fujian, Jiangxi, Shandong, Henan, Hubei, Hunan, Guangdong, Guangxi, Hainan, Sichuan, Guizhou, Yunnan, Shaanxi, Gansu, Ningxia, Xinjiang, and the four direct-administered municipalities are Beijing, Tianjin, Shanghai and Chongqing. A snippet of the total profits for August 2024 is shown in Figure S5, where the map is produced using the R package `hchinamap`. From the estimation of our null model, it is revealed from $\widehat{\boldsymbol{\mu}}$ that Guangdong, Beijing, Jiangsu and Shanghai have significantly larger spatial fixed effects than other provinces or municipalities.

The set of covariates (all standardized) consists of Consumer Price Index (CPI), Purchasing Price Index for industrial producers (PPI) and output of electricity ($elec$). The data is available at the National Bureau of Statistics of China: https://data.stats.gov.cn/english/.

We consider three spatial weight matrix candidates, with each row standardized by its L_1 norm if the row sum exceeds one: inverse distance matrix using inverse of geographical distances between

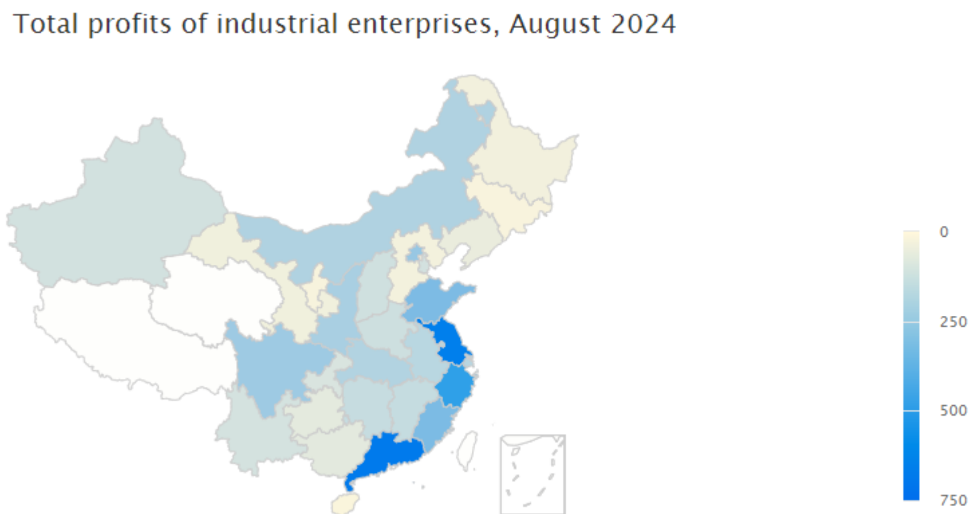

Figure S5: Illustration of the total profits of industrial enterprises within considered Chinese provinces and direct-administered municipalities, in 100 million yuans.

locations computed by the Geodesic WGS-84 System ($\mathbf{W}_1$), contiguity matrix ($\mathbf{W}_2$), municipality matrix such that all direct-administered municipalities are neighbors ($\mathbf{W}_3$).

We treat the covariates as exogenous for two reasons: *CPI* and *PPI* are largely independent of the internal economic activities specific to enterprises within each province or municipality, and electricity supply as a public utility is often price inelastic. Using the aforementioned covariates and spatial weight matrices, we first specify a time-invariant spatial autoregressive model as our null model:

$$profit_t = \boldsymbol{\mu}^* + (\phi^*_{1,0}\mathbf{W}_1 + \phi^*_{2,0}\mathbf{W}_2 + \phi^*_{3,0}\mathbf{W}_3)profit_t + (CPI_t, PPI_t, elec_t)\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t. \tag{S3.1}$$

We estimate the parameters in (S3.1) using our adaptive LASSO estimators. The estimated coefficients $\{\widehat{\phi}_{1,0}, \widehat{\phi}_{2,0}, \widehat{\phi}_{3,0}, \widehat{\boldsymbol{\beta}}\}$ are presented in Table S5, together with the BIC computed according to (4.1). The table also shows the standard errors of $\widehat{\phi}_{1,0}$ and $\widehat{\phi}_{3,0}$ based on Theorem 2. The respective p-values for testing $\widehat{\phi}_{1,0} = 0$ and $\widehat{\phi}_{3,0} = 0$ are both less than 0.0001, revealing some spillovers among the neighbors of provinces and municipalities. Interestingly, $\widehat{\phi}_{3,0}$ suggests a negative spillover effect among the four direct-administered municipalities, which could be explained by that the enterprises within municipalities are main competitors in the market.

Hereafter, we refer to (S3.1) as the null model. The rest of the analysis is performed in an exploratory fashion such that we consider spatial autoregressive models of the form (2.1) with

	$\widehat{\phi}_{1,0}$	$\widehat{\phi}_{2,0}$	$\widehat{\phi}_{3,0}$	$\widehat{\beta}_{CPI}$	$\widehat{\beta}_{PPI}$	$\widehat{\beta}_{elec}$	BIC
Null Model	15.184	.000	-.285	.021	.053	.394	2.790
	(3.725)		(.066)				

Table S5: Estimated coefficients for model (S3.1), with standard errors (in brackets) computed according to the last part of Section 4. $\widehat{\beta}_{CPI}$, $\widehat{\beta}_{PPI}$ and $\widehat{\beta}_{elec}$ denote the estimates of $\boldsymbol{\beta}^*$ corresponding to CPI_t, PPI_t and $elec_t$, respectively.

some l_j and dynamic variables $\{z_{j,k,t}\}$. We consider the following models:

$$\begin{aligned}
&\text{Model 1}: profit_t = \boldsymbol{\mu}^* + \sum_{j=1}^{3}\Big(\phi_{j,0}^* + \sum_{k=1}^{15}\phi_{j,k}^*\mathbb{1}\{t \le 5+5k\}\Big)\mathbf{W}_j profit_t + (CPI_t, PPI_t, elec_t)\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t;\\
&\text{Model 2}: profit_t = \boldsymbol{\mu}^* + \sum_{j=1}^{3}\Big(\phi_{j,0}^* + \sum_{k=1}^{9}\phi_{j,k}^*\mathbb{1}\{\mathrm{sd}(profit_{t-5}) \le \gamma_k\}\Big)\mathbf{W}_j profit_t + (CPI_t, PPI_t, elec_t)\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t,\\
&\qquad \text{where } (\gamma_1,\ldots,\gamma_9) = (.177, .193, .202, .207, .217, .219, .231, .250, .302);\\
&\text{Model 3}: profit_t = \boldsymbol{\mu}^* + \sum_{j=1}^{3}\Big(\phi_{j,0}^* + \sum_{k=1}^{5}\phi_{j,k}^*\mathbb{1}\{t \text{ divides } 2k\}\Big)\mathbf{W}_j profit_t + (CPI_t, PPI_t, elec_t)\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t;\\
&\text{Model 4}: profit_t = \boldsymbol{\mu}^* + \Big(\phi_{1,0}^*\alpha_{3,1}(\mathbf{W}_1) + \phi_{2,0}^*\alpha_{3,2}(\mathbf{W}_1) + \phi_{3,0}^*\alpha_{3,3}(\mathbf{W}_1)\Big) profit_t + (CPI_t, PPI_t, elec_t)\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t.
\end{aligned} \tag{S3.2}$$

Model 1 represents a spatial autoregressive model with the spatial weight matrix potentially changing at $(10, 15, \ldots, 75, 80)$. Model 2 is a self-exciting threshold spatial autoregressive model with the standard deviation of $profit_{t-5}$ as the threshold variable. The sequence of threshold value is in fact the empirical quantile, from 10% to 90%, of $\mathrm{sd}(profit_{t-5})$. Model 3 is similar to the null model but accounts for monthly spillovers for lags of two, four, six, eight and ten months. Model 4 adapts our framework to time-invariant and nonlinear spatial weight matrices, where $\alpha_{3,1}(\mathbf{W}_1)$, $\alpha_{3,2}(\mathbf{W}_1)$ and $\alpha_{3,3}(\mathbf{W}_1)$ denote the matrices formed by series expansion using the order-3 normalized Laguerre functions (inspired by Sun (2016)) based on $\{(\mathbf{W}_1)_{ij}^{-1}\}_{i,j\in[29]}$.

Table S6 reports the estimated parameters and BIC for each model. The nonzero $\widehat{\phi}_{1,9}$ for Model 1 corresponds to a change in the spillovers featured by $\mathbf{W}_1$ in March 2020, potentially suggesting inactive economic activities due to COVID-19 starting at the beginning of 2020.

Model 2 has the best BIC among all models shown here, including the null model. From

	nonzero $\widehat{\phi}_{j,k}$				$\widehat{\beta}_{CPI}$	$\widehat{\beta}_{PPI}$	$\widehat{\beta}_{elec}$	BIC
Model 1	$\widehat{\phi}_{1,9}=18.030$ (13.601)				.044	.038	.309	2.978
Model 2	$\widehat{\phi}_{1,3}=9.721$ (3.487)	$\widehat{\phi}_{1,9}=18.997$ (.717)	$\widehat{\phi}_{3,7}=-.522$ (.001)		.030	.038	.368	2.744
Model 3	$\widehat{\phi}_{1,0}=15.424$ (.000)	$\widehat{\phi}_{2,1}=.335$ (.000)	$\widehat{\phi}_{2,3}=-.331$ (.000)	$\widehat{\phi}_{3,5}=-.401$ (.000)	.023	.052	.398	2.747
Model 4	$\widehat{\phi}_{1,0}=.048$ (.001)	$\widehat{\phi}_{2,0}=-.034$ (.003)	$\widehat{\phi}_{3,0}=.114$ (.001)		.043	.056	.512	2.848

Table S6: Estimated coefficients for different models specified in (S3.2), with standard errors (in brackets) computed according to the last part of Section 4. Refer to Table S5 for the definitions of $\widehat{\beta}_{CPI}$, $\widehat{\beta}_{PPI}$ and $\widehat{\beta}_{elec}$.

Table S6, four threshold regions are identified as

$$\widehat{\mathbf{W}}_t = \begin{cases} 28.718\mathbf{W}_1 - 0.522\mathbf{W}_3, & \mathrm{sd}(\mathit{profit}_{t-5}) \le 0.202; \\ 18.997\mathbf{W}_1 - 0.522\mathbf{W}_3, & 0.202 < \mathrm{sd}(\mathit{profit}_{t-5}) \le 0.231; \\ 18.997\mathbf{W}_1, & 0.231 < \mathrm{sd}(\mathit{profit}_{t-5}) \le 0.302; \\ 0, & \mathrm{sd}(\mathit{profit}_{t-5}) > 0.302. \end{cases}$$

For better illustration, the series of $\widehat{\mathbf{W}}_t$ among Beijing, Shanghai and Guangdong are plotted in Figure S6. We see that the spillovers between Beijing and Shanghai is more significant than their respective spillovers with Guangdong.

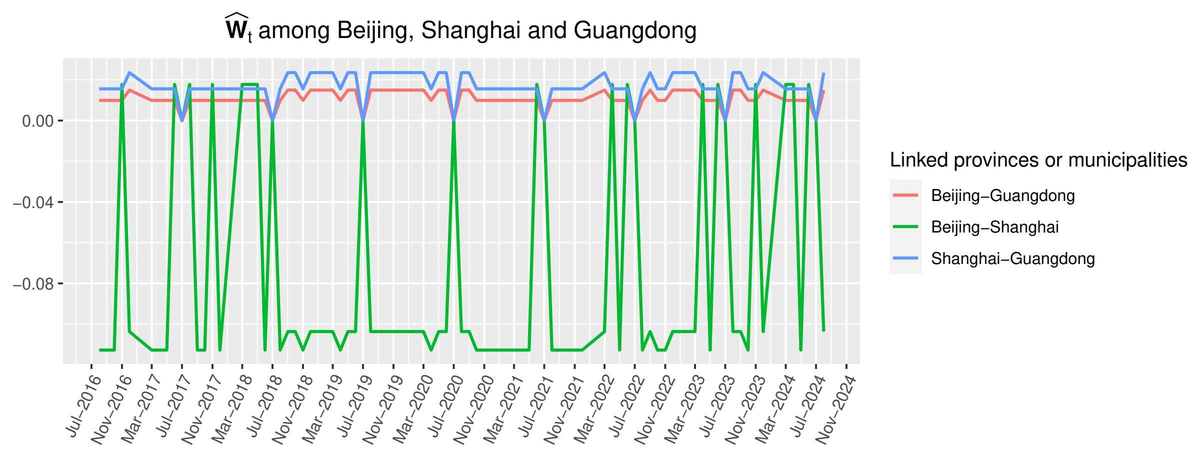

Figure S6: Illustration of $\widehat{\mathbf{W}}_t$ of Model 2 in (S3.2) among Beijing, Shanghai and Guangdong, from August 2016 to August 2024.

On Model 3, Table S6 suggests that the effect of $\mathbf{W}_1$ (representing domestic spillovers) remains constant, that of $\mathbf{W}_2$ (representing more local spillovers) persists every two months but roughly cancels out every half year, and the "municipality spillover" by $\mathbf{W}_3$ occurs every December. The various spillover patterns featured by the expert spatial weight matrices are intriguing and warrant further investigation.

Model 4 considers a time-invariant spillover effect. An example of the estimated spatial weight matrix is displayed in Figure S7. It depicts how the spillovers diminish with the geographical distance. Lastly, the analysis on the total profits data serves to demonstrate our spatial autoregressive framework. More comprehensive investigations are required to further understand the spatial relations among industrial enterprises in Chinese provinces and direct-administered municipalities.

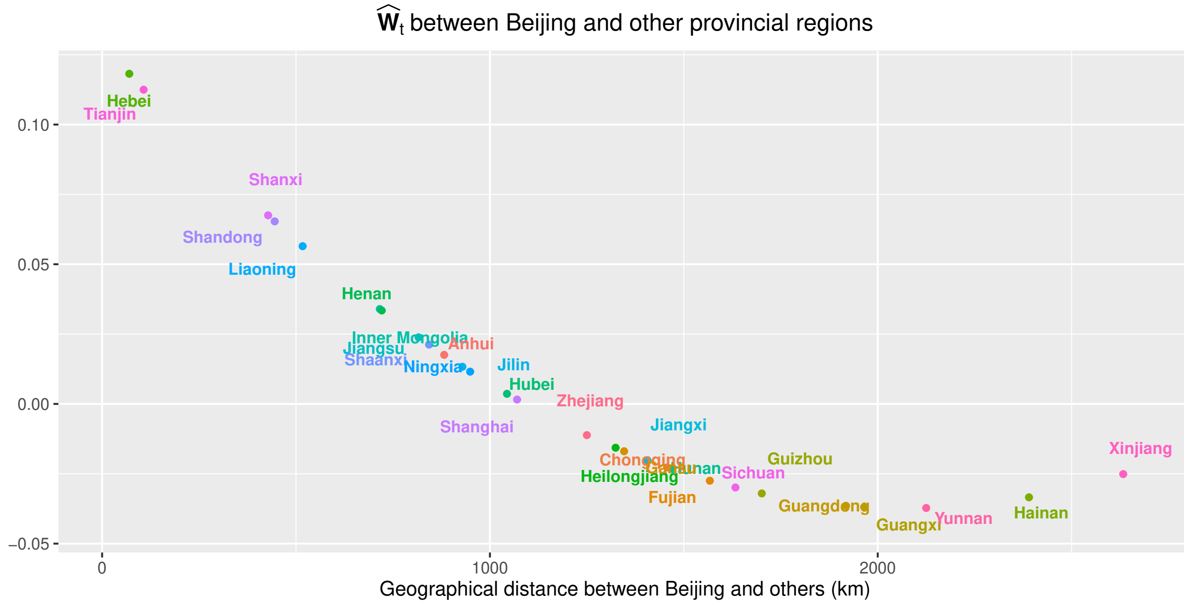

Figure S7: Illustration of (time-invariant) $\widehat{\mathbf{W}}_t$ of Model 4 in (S3.2) between Beijing and other provincial regions, against their geographical distances.

S4. PROOFS OF THEOREMS AND AUXILIARY RESULTS

S4.1. Lemmas and their proofs

To prove our main theorems, recall first $B_{t,ij}$ and $X_{t,ij}$ represent the (i,j) entry of $\mathbf{B}_t$ and $\mathbf{X}_t$ respectively. Define $\mathcal{M}=\bigcap_{i=1}^{13}\mathcal{A}_i$, where

$$
\begin{aligned}
\mathcal{A}_1 &= \left\{ \max_{i,q\in[d]}\max_{j,l\in[r]} \left| \frac{1}{T}\sum_{t=1}^{T}[B_{t,ij}X_{t,ql} - \mathbb{E}(B_{t,ij}X_{t,ql})] \right| < c_T \right\},\\
\mathcal{A}_2 &= \left\{ \max_{i,q\in[d]}\max_{j\in[r]} \left| \frac{1}{T}\sum_{t=1}^{T} B_{t,ij}\epsilon_{t,q} \right| < c_T \right\},\\
\mathcal{A}_3 &= \left\{ \max_{j\in[r]} \left| \frac{1}{T}\sum_{t=1}^{T}\sum_{q=1}^{d} B_{t,qj}\epsilon_{t,q} \right| < c_T d^{\frac{1}{2}+\frac{1}{2w}} \right\},\\
\mathcal{A}_4 &= \left\{ \max_{i\in[d]}\max_{j\in[r]} \left| \bar{B}_{\cdot,ij} - \mathbb{E}[B_{t,ij}] \right| < c_T \right\},\\
\mathcal{A}_5 &= \left\{ \max_{q\in[d]} \left| \bar{\epsilon}_{\cdot,q} \right| < c_T \right\},\\
\mathcal{A}_6 &= \left\{ \max_{i\in[d]}\max_{j\in[r]} \left| \bar{X}_{\cdot,ij} \right| < c_T \right\},\\
\mathcal{A}_7 &= \left\{ \max_{j\in[r]} \left| \sum_{q=1}^{d} \bar{B}_{\cdot,qj}\bar{\epsilon}_{\cdot,q} \right| < 2^{1/2} c_T d^{1/2} \log^{1/2}(T\vee d) S_\epsilon(\mu_{b,\max} + c_T) \right\}, \qquad \text{(S4.1)}\\
\mathcal{A}_8 &= \left\{ \max_{m\in[p]}\max_{n\in[l_m]\cup\{0\}}\max_{i,q\in[d]}\max_{j\in[v]}\max_{l\in[r]} \left| \frac{1}{T}\sum_{t=1}^{T}[z_{m,n,t}B_{t,ij}X_{t,ql} - \mathbb{E}(z_{m,n,t}B_{t,ij}X_{t,ql})] \right| < c_T \right\},\\
\mathcal{A}_9 &= \left\{ \max_{m\in[p]}\max_{n\in[l_m]\cup\{0\}}\max_{i\in[d]}\max_{j\in[r]} \left| \frac{1}{T}\sum_{t=1}^{T} z_{m,n,t}X_{t,ij} \right| < c_T \right\},\\
\mathcal{A}_{10} &= \left\{ \max_{m\in[p]}\max_{n\in[l_m]\cup\{0\}}\max_{i,q\in[d]}\max_{j\in[v]} \left| \frac{1}{T}\sum_{t=1}^{T} z_{m,n,t}B_{t,ij}\epsilon_{t,q} \right| < c_T \right\},\\
\mathcal{A}_{11} &= \left\{ \max_{m\in[p]}\max_{n\in[l_m]\cup\{0\}}\max_{q\in[d]} \left| \frac{1}{T}\sum_{t=1}^{T} z_{m,n,t}\epsilon_{t,q} \right| < c_T \right\},\\
\mathcal{A}_{12} &= \left\{ \max_{m\in[p]}\max_{n\in[l_m]\cup\{0\}}\max_{i\in[d]}\max_{j\in[v]} \left| \frac{1}{T}\sum_{t=1}^{T}[z_{m,n,t}B_{t,ij} - \mathbb{E}(z_{m,n,t}B_{t,ij})] \right| < c_T \right\},\\
\mathcal{A}_{13} &= \left\{ \max_{m\in[p]}\max_{n\in[l_m]\cup\{0\}} \left| \frac{1}{T}\sum_{t=1}^{T} z_{m,n,t} \right| < c_T \vee z_{\max} \right\},
\end{aligned}
$$

where $\bar{B}_{\cdot,ij} := T^{-1}\sum_{t=1}^{T} B_{t,ij}$, $\bar{X}_{\cdot,ij} := T^{-1}\sum_{t=1}^{T} X_{t,ij}$, $\bar{\epsilon}_{\cdot,q} := T^{-1}\sum_{t=1}^{T}\epsilon_{t,q}$, $\mu_{b,\max} := \max_{i,j}\left|\mathbb{E}[B_{t,ij}]\right|$ being a constant implied by Assumption (M1), and $z_{\max}$ is the upper bound for $\{z_{j,k,t}\}$ implied in (M2) with $z_{\max} = 1$ for $\{z_{j,0,t}\}$ by default. Our main theoretical results depict the properties of estimators on the set $\mathcal{M}$ which holds with probability approaching 1 as $T,d \to \infty$ by Assumption (R10), as shown in Lemma S2 which is similar to Theorem S.1 of Lam and Souza (2020).

To prove Lemma S2, we first quote a Nagaev-type inequality for functional dependent data from Theorem 2(ii), 2(iii) and Section 4 of Liu et al. (2013), presented as the following lemma.

Lemma S1 *For a zero mean time series process $\mathbf{x}_t = \mathbf{f}(\mathcal{F}_t)$ defined in (3.1) with dependence measure $\theta^{x}_{t,q,i}$ defined in (3.2), assume $\Theta^{x}_{m,2w} \le Cm^{-\alpha}$ as in Assumption (M1). Then there exists constants C_1, C_2 and C_3 independent of n,T and the index i such that*

$$\mathbb{P}\left(\left|\frac{1}{T}\sum_{t=1}^{T} x_{t,i}\right| > n\right) \le \frac{C_1 T^{w(1/2-\widetilde{\alpha})}}{(Tn)^{w}} + C_2 \exp(-C_3 T^{\widetilde{\beta}} n^2),$$

where $\widetilde{\alpha} = \alpha \wedge (1/2 - 1/w)$ and $\widetilde{\beta} = (3 + 2\widetilde{\alpha} w)/(1+w)$.

Furthermore, assume another zero mean time series process $\{\mathbf{e}_t\}$ (can be the same process $\{\mathbf{x}_t\}$) with $\Theta^{e}_{m,2w}$ as in (M1). Then provided $\max_i \|x_{t,i}\|_{2w}, \max_j \|e_{t,j}\|_{2w} \le c_0 < \infty$ where c_0 is a constant, the above Nagaev-type inequality holds for the product process $\{x_{t,i}e_{t,j} - \mathbb{E}(x_{t,i}e_{t,j})\}$.

The above results also hold for any zero mean non-stationary process $\mathbf{x}_t = \mathbf{f}_t(\mathcal{F}_t)$ provided that $\max_i \|x_{t,i}\|_w < \infty$ and $\Theta^{x,*}_{m,2w} \le Cm^{-\alpha}$, where $\Theta^{x,*}_{m,q}$ is uniform tail sum defined as

$$\Theta^{x,*}_{m,q} := \sum_{t=m}^{\infty} \max_i \theta^{x,*}_{t,q,i} := \sum_{t=m}^{\infty} \max_i \sup_t \left\|x_{t,i} - x'_{t,i}\right\|_q .$$

We present Lemma S2 below. Note that we assume $\alpha > 1/2 - 1/w$ which can be relaxed at the cost of more complicated rates and longer proofs presented here, and it simplifies the form of Lemma S1 as $w(1/2 - \widetilde{\alpha}) = \widetilde{\beta} = 1$.

Lemma S2 *Let Assumptions (M1), (M2) (or (M2')), and (R2) hold and $\alpha > 1/2 - 1/w$ in Assumption (M1). Suppose for the application of the Nagaev-type inequality in Lemma S1 for the processes in $\mathcal{M} = \bigcap_{i=1}^{13} \mathcal{A}_i$ where $\mathcal{A}_i$ is defined in (S4.1), the constants C_1, C_2 and C_3 are the same. Then with $g \ge \sqrt{3/C_3}$ where g is the constant defined in $c_T = gT^{-1/2}\log^{1/2}(T \vee d)$, we have*

$$\mathbb{P}(\mathcal{M}) \ge 1 - 14C_1 r^2 vL \left(\frac{C_3}{3}\right)^{w/2} \frac{d^2}{T^{w/2-1}\log^{w/2}(T\vee d)} - \frac{14C_2 r^2 vLd^2}{T^3 \vee d^3} - \frac{2r}{T \vee d}.$$

Proof of Lemma S2. As shown in Theorem S.1 of Lam and Souza (2020), the tail condition in Assumption (M1) implies $\|\cdot\|_{2w}$ is bounded for the processes $\{B_{t,ij}X_{t,ql} - \mathbb{E}(B_{t,ij}X_{t,ql})\}$, $\{B_{t,ij}\epsilon_{t,q}\}$, $\{X_{t,ij}\}$ and $\{\epsilon_{t,q}\}$, and further with Assumption (R2) we have

$$
\begin{aligned}
\mathbb{P}(\mathcal{A}_1^c) &\le C_1 r^2 \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d^2}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 r^2 d^2}{T^3 \vee d^3},\\
\mathbb{P}(\mathcal{A}_2^c) &\le C_1 r \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d^2}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 r d^2}{T^3 \vee d^3},\\
\mathbb{P}(\mathcal{A}_3^c) &\le C_1 \Big(\frac{C_3}{3}\Big)^{w/2} \frac{r}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 r}{T^3 \vee d^3},\\
\mathbb{P}(\mathcal{A}_4^c) &\le C_1 r \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 r d}{T^3 \vee d^3},\\
\mathbb{P}(\mathcal{A}_5^c) &\le C_1 \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 d}{T^3 \vee d^3},\\
\mathbb{P}(\mathcal{A}_6^c) &\le C_1 r \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 r d}{T^3 \vee d^3},\\
\mathbb{P}(\mathcal{A}_7^c) &\le \frac{2r}{T\vee d} + \mathbb{P}(\mathcal{A}_4^c) + \mathbb{P}(\mathcal{A}_5^c).
\end{aligned}
$$

Consider the remaining sets. First let Assumption (M2) hold and it is easy to see $\|\cdot\|_{2w}$ is bounded for the processes $\{z_{m,n,t}B_{t,ij}X_{t,ql} - \mathbb{E}(z_{m,n,t}B_{t,ij}X_{t,ql})\}$, $\{z_{m,n,t}B_{t,ij}\epsilon_{t,q}\}$, $\{z_{m,n,t}X_{t,ij}\}$ and $\{z_{m,n,t}\epsilon_{t,q}\}$, and their uniform tail sums satisfy the condition in Lemma S1. Thus, apply Lemma S1 first on $\mathcal{A}_8^c$ and we have by the union bound,

$$
\begin{aligned}
\mathbb{P}(\mathcal{A}_8^c) &\le \sum_{m\in[p]}\sum_{n\in[l_m]}\sum_{i,q\in[d]}\sum_{j\in[v]}\sum_{l\in[r]} \mathbb{P}\left(\left| \frac{1}{T}\sum_{t=1}^{T} [z_{m,n,t}B_{t,ij}X_{t,ql} - \mathbb{E}(z_{m,n,t}B_{t,ij}X_{t,ql})] \right| \ge c_T \right)\\
&\le d^2 rvL\Big(\frac{C_1 T}{(Tc_T)^w} + C_2 \exp(-C_3 T c_T^2) \Big)\\
&\le C_1 rvL \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d^2}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 rvL d^2}{T^3 \vee d^3}.
\end{aligned}
$$

Similarly using Lemma S1, we have

$$
\begin{aligned}
\mathbb{P}(\mathcal{A}_9^c) &\le C_1 rL \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 rLd}{T^3 \vee d^3},\\
\mathbb{P}(\mathcal{A}_{10}^c) &\le C_1 vL \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d^2}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 vLd^2}{T^3 \vee d^3},\\
\mathbb{P}(\mathcal{A}_{11}^c) &\le C_1 L \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 Ld}{T^3 \vee d^3},
\end{aligned}
$$

$$\mathbb{P}(\mathcal{A}_{12}^c) \le C_1 v L \Big(\frac{C_3}{3}\Big)^{w/2} \frac{d}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 v L d}{T^3 \vee d^3},$$

while $\mathbb{P}(\mathcal{A}_{13}^c)=0$ by Assumption (M2). On the other hand, if we have Assumption (M2'), the above results remain valid, except that

$$\mathbb{P}(\mathcal{A}_{13}^c) \le C_1 L \Big(\frac{C_3}{3}\Big)^{w/2} \frac{1}{T^{w/2-1}\log^{w/2}(T\vee d)} + \frac{C_2 L}{T^3 \vee d^3},$$

by applying Lemma S1 given the bounded support and tail sum assumption in (M2'). For any $\{z_{j,0,t}\}$, we may treat it as a non-stochastic basis as in (M2) and the result follows. Lastly, by $\mathbb{P}(\mathcal{M}) \ge 1 - \sum_{i=1}^{13} \mathbb{P}(\mathcal{A}_i^c)$ we complete the proof of Lemma S2. □

We present the following lemma with a short proof as well, and we will frequently utilize the defined notation $\mathbf{V}_{\mathbf{H},K}$ (which is the same definition in Theorem 2) in the proof of main theorems.

Lemma S3 *For any $n\times d$ matrix $\mathbf{H}=(\mathbf{h}_1,\ldots,\mathbf{h}_n)^\top$ and any $d\times K$ matrix $\mathbf{M}$, define*

$$\mathbf{V}_{\mathbf{H},K} = \begin{pmatrix} \mathbf{I}_K \otimes \mathbf{h}_1 \\ \vdots \\ \mathbf{I}_K \otimes \mathbf{h}_n \end{pmatrix}.$$

We then have $\mathbf{H}\mathbf{M} = \Big(\mathbf{I}_n \otimes \mathrm{vec}(\mathbf{M})^\top\Big)\mathbf{V}_{\mathbf{H},K}$.

Proof of Lemma S3. Notice that

$$\Big(\mathbf{I}_n \otimes \mathrm{vec}(\mathbf{M})^\top\Big)\mathbf{V}_{\mathbf{H},K} = \begin{pmatrix} \mathrm{vec}\big(\mathbf{M}\big)^\top (\mathbf{I}_K \otimes \mathbf{h}_1) \\ \vdots \\ \mathrm{vec}\big(\mathbf{M}\big)^\top (\mathbf{I}_K \otimes \mathbf{h}_n) \end{pmatrix}.$$

The j-th row of this matrix (as a column vector) is hence

$$(\mathbf{I}_K \otimes \mathbf{h}_j^\top)\mathrm{vec}\big(\mathbf{M}\big) = \mathrm{vec}\big(\mathbf{h}_j^\top \mathbf{M}\big) = \mathrm{vec}\big(\mathbf{M}^\top \mathbf{h}_j\big) = \mathbf{M}^\top \mathbf{h}_j,$$

which is the j-th row of $\mathbf{H}\mathbf{M}$ indeed. □

Remark S1 *With the notation of $\mathbf{V}_{\mathbf{H},K}$, we may write any $d\times K$ matrix $\mathbf{M}$ as*

$$\mathbf{M} = \mathbf{I}_d \mathbf{M} = \Big(\mathbf{I}_d \otimes \mathrm{vec}(\mathbf{M})^\top\Big)\mathbf{V}_{\mathbf{I}_d,K},$$

which will be useful if we are interested in the interaction only between $\mathbf{A},\mathbf{M}$ *in* $\mathbf{ABM}$*, with* $\mathbf{A}\in\mathbb{R}^{l\times r}$ *and* $\mathbf{B}\in\mathbb{R}^{r\times d}$*, since we have*

$$\mathbf{ABM}=(\mathbf{B}^\top\mathbf{A}^\top)^\top\mathbf{M}=\mathbf{V}_{\mathbf{B}^\top,l}^\top\Big(\mathbf{I}_d\otimes\mathrm{vec}(\mathbf{A}^\top)\Big)\mathbf{M}=\mathbf{V}_{\mathbf{B}^\top,l}^\top\Big(\mathbf{I}_d\otimes\mathrm{vec}(\mathbf{A}^\top)\mathrm{vec}(\mathbf{M})^\top\Big)\mathbf{V}_{\mathbf{I}_d,K}.$$

Moreover, notice that in Lemma S3, if $K=1$*, i.e.* $\mathbf{M}$ *is a vector, then* $\mathbf{V}_{\mathbf{H},1}=\mathrm{vec}(\mathbf{H}^\top)$ *and Lemma S3 simply coincides with the fact that*

$$\mathbf{BM}=\mathrm{vec}(\mathbf{M}^\top\mathbf{B}^\top)=\mathrm{vec}(\mathbf{M}^\top\mathbf{B}^\top\mathbf{I}_r)=(\mathbf{I}_r\otimes\mathbf{M}^\top)\mathrm{vec}(\mathbf{B}^\top)=\Big(\mathbf{I}_r\otimes\mathrm{vec}(\mathbf{M})^\top\Big)\mathbf{V}_{\mathbf{B},1}.$$

Thus, we can interpret $\mathbf{V}_{\mathbf{H},K}$ *to be a "K-block vectorization" of* $\mathbf{H}$*, as a generalization of vectorization.*

Lemma S4 *Let the assumptions in Theorem 2 hold. Let* $\mathbf{R}_\beta$ *and* $\mathbf{\Sigma}_\beta$ *be defined in Theorem 2. For* $I_2=\big[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\big]^{-1}T^{-2}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu$*, we have* I_2 *asymptotically normal with rate* $T^{-1/2}d^{-(1-b)/2}$ *such that*

$$T^{1/2}(\mathbf{R}_\beta\mathbf{\Sigma}_\beta\mathbf{R}_\beta^\top)^{-1/2}I_2\xrightarrow{D}\mathcal{N}(\mathbf{0},\mathbf{I}_r).$$

Proof of Lemma S4. Given any non-zero $\boldsymbol{\alpha}\in\mathbb{R}^r$ with $\|\boldsymbol{\alpha}\|_1\le c<\infty$, we construct below the asymptotic normality of $\boldsymbol{\alpha}^\top I_2$ which is $T^{1/2}d^{(1-b)/2}$-convergent. First, we can further decompose

$$\begin{aligned}\boldsymbol{\alpha}^\top I_2=&\big[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\big]^{-1}\big(T^{-1}\mathbf{X}^\top\mathbf{B}^\nu-\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\big)T^{-1}(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu\\&+\big[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\big]^{-1}\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)T^{-1}(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu,\end{aligned}$$

with the second term dominating the first by (S4.6). Recall that

$$\mathbf{R}_\beta=\big[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\big]^{-1}\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t),$$

then

$$\boldsymbol{\alpha}^\top I_2=\frac{1}{T}\sum_{t=1}^T\boldsymbol{\alpha}^\top\mathbf{R}_\beta(\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t))^\top\boldsymbol{\epsilon}_t(1+o_P(1)).$$

To construct the asymptotic normality of $\boldsymbol{\alpha}^\top I_2$, we want to show the following as in (S4.33) that

$$\sum_{t\ge0}\Big\|P_0(\boldsymbol{\alpha}^\top\mathbf{R}_\beta(\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t))^\top\boldsymbol{\epsilon}_t)\Big\|_2<\infty,\tag{S4.2}$$

so that Theorem 3 (ii) of Wu (2011) can then be applied. With the definition $s_2 := \boldsymbol{\alpha}^\top \mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top \boldsymbol{\alpha}$, we have $T^{1/2} s_2^{-1/2} \boldsymbol{\alpha}^\top I_2 \xrightarrow{D} \mathcal{N}(0,1)$ and hence

$$T^{1/2}(\mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top)^{-1/2} I_2 \xrightarrow{D} \mathcal{N}(\mathbf{0}, \mathbf{I}_r). \tag{S4.3}$$

Similar to the proof of (S4.33), we have (S4.2) hold by Assumption (R7) and

$$\begin{aligned}
&\Big\| P_0(\boldsymbol{\alpha}^\top \mathbf{R}_\beta (\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top \boldsymbol{\epsilon}_t) \Big\|_2 \\
&= \Big\| \boldsymbol{\alpha}^\top \mathbf{R}_\beta \big\{ P_0((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top) \mathbb{E}_0(\boldsymbol{\epsilon}_t) \big\} + \boldsymbol{\alpha}^\top \mathbf{R}_\beta \big\{ \mathbb{E}_{-1}((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top) P_0(\boldsymbol{\epsilon}_t) \big\} \Big\|_2 \\
&\le \Big\{ 2\boldsymbol{\alpha}^\top \mathbf{R}_\beta \mathbb{E} \Big\{ P_0((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top) \mathbb{E}_0(\boldsymbol{\epsilon}_t) \mathbb{E}_0(\boldsymbol{\epsilon}_t^\top) P_0(\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)) \Big\} \mathbf{R}_\beta^\top \boldsymbol{\alpha} \Big\}^{1/2} \\
&\quad + \Big\{ 2\boldsymbol{\alpha}^\top \mathbf{R}_\beta \mathbb{E} \Big\{ \mathbb{E}_{-1}((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top) P_0(\boldsymbol{\epsilon}_t) P_0(\boldsymbol{\epsilon}_t^\top) \mathbb{E}_{-1}(\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)) \Big\} \mathbf{R}_\beta^\top \boldsymbol{\alpha} \Big\}^{1/2} \\
&= O\Big(\big\|\boldsymbol{\alpha}\big\|_1 \big\|\mathbf{R}_\beta\big\|_\infty \Big) \Big(d \cdot \max_{j \in [d]} \mathbb{E}^{1/2}(\mathbb{E}_0^2(\epsilon_{t,j})) \max_{j \in [d]} \max_{k \in [v]} \big\| P_0(B_{t,jk}) \big\|_2 + d \cdot \sigma_{\max} \max_{j \in [d]} \big\| P_0(\epsilon_{t,j}) \big\|_2 \Big) \\
&= O\Big(\max_{j \in [d]} \big\| P_0^\epsilon(\epsilon_{t,j}) \big\|_2 + \max_{j \in [d]} \max_{k \in [v]} \big\| P_0^b(B_{t,jk}) \big\|_2 \Big),
\end{aligned}$$

where the second last equality used Assumption (R2), and the last used $\big\|\mathbf{R}_\beta\big\|_\infty = O(d^{-2} \cdot d) = O(d^{-1})$ by (S4.4) and Assumption (R3).

It remains to show $\boldsymbol{\alpha}^\top I_2$ is indeed of order $T^{-1/2} d^{-(1-b)/2}$. To this end, we only need to show s_2 is of order $d^{-(1-b)}$. First, we have $\mathbf{R}_\beta \mathbf{R}_\beta^\top = \big[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)\big]^{-1}$ which has all eigenvalues of order d^{-2} from (S4.4) and Assumption (R3). Consider any j-th diagonal element of $\boldsymbol{\Sigma}_\beta$, we have

$$\begin{aligned}
(\boldsymbol{\Sigma}_\beta)_{jj} &= \sum_\tau \mathbb{E} \big\{ (\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))_{\cdot j}^\top \boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_{t+\tau}^\top (\mathbf{B}_{t+\tau} - \mathbb{E}(\mathbf{B}_t))_{\cdot j} \big\} \\
&= \sum_\tau \mathrm{tr} \big\{ \mathbb{E}((\mathbf{B}_{t+\tau} - \mathbb{E}(\mathbf{B}_t))_{\cdot j} (\mathbf{B}_{t+\tau} - \mathbb{E}(\mathbf{B}_t))_{\cdot j}^\top) \mathbb{E}(\boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_{t+\tau}^\top) \big\},
\end{aligned}$$

which is finite and has order exactly d^{1+b} by Assumption (R8). Notice $\boldsymbol{\Sigma}_\beta$ is $r \times r$, the order of eigenvalues of $\boldsymbol{\Sigma}_\beta$ is hence exactly d^{1+b}. The order of s_2 is d^{b-1} by

$$\|\boldsymbol{\alpha}\|_1^2 \lambda_{\min}(\mathbf{R}_\beta \mathbf{R}_\beta^\top) \lambda_{\min}(\boldsymbol{\Sigma}_\beta) \le s_2 \le \|\boldsymbol{\alpha}\|_1^2 \lambda_{\max}(\mathbf{R}_\beta \mathbf{R}_\beta^\top) \lambda_{\max}(\boldsymbol{\Sigma}_\beta).$$

This completes the proof of Lemma S4. □

S4.2. Proofs of corollaries

Proof of Corollary 1. This is direct from Theorem 2. □

Proof of Corollary S1. This is direct from Theorem 2. □

Proof of Corollary 2. This is direct from Theorem 2. □

Proof of Corollary 3. This is direct from Theorem 2. □

S4.3. Proofs of theorems

Proof of Theorem 1. From (2.12) and (2.13), we have

$$\begin{aligned}
\boldsymbol{\beta}(\boldsymbol{\phi}^*) &= \left(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\left(\mathbf{y}^\nu - \sum_{j=1}^{p}\sum_{k=0}^{l_j}\phi^*_{j,k}\mathbf{W}^\otimes_j\mathbf{y}^\nu_{j,k}\right)\\
&= \left(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\left(\mathbf{1}_T\otimes\boldsymbol{\mu}^* + \mathbf{X}\boldsymbol{\beta}^* + \boldsymbol{\epsilon}^\nu\right)\\
&= \boldsymbol{\beta}^* + \left(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu.
\end{aligned}$$

We now define a diagonal matrix $\mathbf{D}_{z_{j,k}} := \operatorname{diag}(z_{j,k,1}\mathbf{I}_d, \ldots, z_{j,k,T}\mathbf{I}_d) \in \mathbb{R}^{dT\times dT}$, with diagonal blocks $z_{j,k,1}\mathbf{I}_d, \ldots, z_{j,k,T}\mathbf{I}_d$, and $\boldsymbol{\Pi}^{*\otimes} := \left(\mathbf{I}_{Td} - \sum_{j=1}^{p}\sum_{k=0}^{l_j}\phi^*_{j,k}\mathbf{W}^\otimes_j\mathbf{D}_{z_{j,k}}\right)^{-1}$. We then have $\mathbf{y}^\nu = \boldsymbol{\Pi}^{*\otimes}\left(\mathbf{1}_T\otimes\boldsymbol{\mu}^* + \mathbf{X}\boldsymbol{\beta}^* + \boldsymbol{\epsilon}^\nu\right)$. Thus,

$$\begin{aligned}
\boldsymbol{\beta}(\widetilde{\boldsymbol{\phi}}) &= \left(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\left(\mathbf{y}^\nu - \sum_{j=1}^{p}\sum_{k=0}^{l_j}\widetilde{\phi}_{j,k}\mathbf{W}^\otimes_j\mathbf{y}^\nu_{j,k}\right)\\
&= \boldsymbol{\beta}(\boldsymbol{\phi}^*) + \left(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\left(\sum_{j=1}^{p}\sum_{k=0}^{l_j}(\phi^*_{j,k} - \widetilde{\phi}_{j,k})\mathbf{W}^\otimes_j\mathbf{y}^\nu_{j,k}\right)\\
&= \boldsymbol{\beta}(\boldsymbol{\phi}^*) + \left(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\\
&\qquad\cdot\left(\sum_{j=1}^{p}\sum_{k=0}^{l_j}(\phi^*_{j,k} - \widetilde{\phi}_{j,k})\mathbf{W}^\otimes_j\mathbf{D}_{z_{j,k}}\boldsymbol{\Pi}^{*\otimes}\left(\mathbf{1}_T\otimes\boldsymbol{\mu}^* + \mathbf{X}\boldsymbol{\beta}^* + \boldsymbol{\epsilon}^\nu\right)\right).
\end{aligned}$$

We can hence decompose $\boldsymbol{\beta}(\widetilde{\boldsymbol{\phi}}) - \boldsymbol{\beta}^* = \sum_{j=1}^{5} I_j$ where

$$\begin{aligned}
I_1 &:= \left[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\right]^{-1}\left[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t) - T^{-2}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\right](\boldsymbol{\beta}(\widetilde{\boldsymbol{\phi}}) - \boldsymbol{\beta}^*),\\
I_2 &:= \left[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\right]^{-1}T^{-2}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu,
\end{aligned}$$

$$I_3 := \left[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)\right]^{-1} T^{-2}\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \sum_{j=1}^{p}\sum_{k=0}^{l_j}(\phi_{j,k}^* - \widetilde{\phi}_{j,k})\mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes}\mathbf{X}\boldsymbol{\beta}^*,$$

$$I_4 := \left[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)\right]^{-1} T^{-2}\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \sum_{j=1}^{p}\sum_{k=0}^{l_j}(\phi_{j,k}^* - \widetilde{\phi}_{j,k})\mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes}\boldsymbol{\epsilon}^\nu,$$

$$I_5 := \left[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)\right]^{-1} T^{-2}\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \sum_{j=1}^{p}\sum_{k=0}^{l_j}(\phi_{j,k}^* - \widetilde{\phi}_{j,k})\mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes}(\mathbf{1}_T \otimes \boldsymbol{\mu}^*).$$

Notice we can take any $t \in [T]$ for $\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)$ and $\mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)$ due to Assumption (M1). To bound I_1 to I_4, we first have

$$\left\| \left[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)\right]^{-1} \right\|_1 \le \frac{r^{1/2}}{\lambda_r\left[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)\right]} \le \frac{r^{1/2}}{d^2 u^2}, \tag{S4.4}$$

where $\lambda_r\left[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)\right] = \sigma_r^2(\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)) \ge d^2u^2$ with $u > 0$ being a constant by Assumption (R3), and $\lambda_j(\cdot)$ and $\sigma_j(\cdot)$ denoting the j-th largest eigenvalue and singular value of a matrix respectively.

Next, define

$$\mathbf{U} = \mathbf{I}_d \otimes T^{-1}\sum_{t=1}^{T}\mathrm{vec}\big(\mathbf{B}_t - \bar{\mathbf{B}}\big)\mathbf{x}_t^\top, \ \ \mathbf{U}_0 = \mathbf{I}_d \otimes \mathbb{E}(\mathbf{b}_t\mathbf{x}_t^\top).$$

By Lemma S3, we then have

$$\begin{aligned}
T^{-1}\mathbf{X}^\top \mathbf{B}^v &= T^{-1}\sum_{t=1}^{T}\mathbf{X}_t^\top(\mathbf{B}_t - \bar{\mathbf{B}}) \\
&= T^{-1}\sum_{t=1}^{T}\Big\{\big(\mathbf{I}_d \otimes \mathbf{x}_t^\top\big)\mathbf{V}_{\mathbf{I}_d,r}\Big\}^\top \Big\{\big(\mathbf{I}_d \otimes \mathrm{vec}\big(\mathbf{B}_t - \bar{\mathbf{B}}\big)^\top\big)\mathbf{V}_{\mathbf{I}_d,v}\Big\} \\
&= T^{-1}\sum_{t=1}^{T}\mathbf{V}_{\mathbf{I}_d,r}^\top\big(\mathbf{I}_d \otimes \mathbf{x}_t\big)\big(\mathbf{I}_d \otimes \mathrm{vec}\big(\mathbf{B}_t - \bar{\mathbf{B}}\big)^\top\big)\mathbf{V}_{\mathbf{I}_d,v} = \mathbf{V}_{\mathbf{I}_d,r}^\top \mathbf{U}^\top \mathbf{V}_{\mathbf{I}_d,v}.
\end{aligned}$$

Similarly we have $\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) = \mathbf{V}_{\mathbf{I}_d,r}^\top \mathbf{U}_0^\top \mathbf{V}_{\mathbf{I}_d,v}$. Thus on the set $\mathcal{M}$ in Lemma S2, with (S4.4) it holds that

$$
\begin{aligned}
&\left\|I_1\right\|_1 \leq \frac{r^{1/2}}{d^2u^2}\cdot\left\|\mathbf{V}_{\mathbf{I}_d,r}^\top\mathbf{U}_0^\top\mathbf{V}_{\mathbf{I}_d,v}\mathbf{V}_{\mathbf{I}_d,v}^\top\mathbf{U}_0-\mathbf{V}_{\mathbf{I}_d,r}^\top\mathbf{U}^\top\mathbf{V}_{\mathbf{I}_d,v}\mathbf{V}_{\mathbf{I}_d,v}^\top\mathbf{U}\right\|_1\cdot\left\|\mathbf{V}_{\mathbf{I}_d,r}(\boldsymbol{\beta}(\widetilde{\boldsymbol{\phi}})-\boldsymbol{\beta}^*)\right\|_1\\
&=O\Big(\frac{1}{d}\Big)\Big(\left\|\mathbf{V}_{\mathbf{I}_d,r}^\top(\mathbf{U}_0-\mathbf{U})^\top\mathbf{V}_{\mathbf{I}_d,v}\mathbf{V}_{\mathbf{I}_d,v}^\top\mathbf{U}_0\right\|_1+\left\|\mathbf{V}_{\mathbf{I}_d,r}^\top\mathbf{U}^\top\mathbf{V}_{\mathbf{I}_d,v}\mathbf{V}_{\mathbf{I}_d,v}^\top(\mathbf{U}_0-\mathbf{U})\right\|_1\Big)\left\|\boldsymbol{\beta}(\widetilde{\boldsymbol{\phi}})-\boldsymbol{\beta}^*\right\|_1\\
&=O\Big(\frac{1}{d}\Big)\Big\{d\cdot\left\|\mathbf{U}_0-\mathbf{U}\right\|_{\max}\cdot\left\|\mathbf{U}_0\right\|_{\max}\\
&\quad+d\cdot\left\|\mathbf{U}_0-\mathbf{U}\right\|_{\max}\cdot\Big(\left\|\mathbf{U}_0-\mathbf{U}\right\|_{\max}+\left\|\mathbf{U}_0\right\|_{\max}\Big)\Big\}\cdot\left\|\boldsymbol{\beta}(\widetilde{\boldsymbol{\phi}})-\boldsymbol{\beta}^*\right\|_1=O\Big(c_T\left\|\boldsymbol{\beta}(\widetilde{\boldsymbol{\phi}})-\boldsymbol{\beta}^*\right\|_1\Big),
\end{aligned}
\tag{S4.5}
$$

where the last equality used Assumption (R3) which implies $\left\|\mathbf{U}_0\right\|_{\max}$ is bounded by some constant, and Lemma S2 (using $\mathcal{A}_1,\mathcal{A}_4,\mathcal{A}_6$) that

$$
\begin{aligned}
&\left\|\mathbf{U}-\mathbf{U}_0\right\|_{\max}=\left\|\frac{1}{T}\sum_{t=1}^T\big[\mathrm{vec}\big(\mathbf{B}_t-\bar{\mathbf{B}}\big)\mathbf{x}_t^\top\big]-\mathbb{E}(\mathbf{b}_t\mathbf{x}_t^\top)\right\|_{\max}\\
&=\left\|\frac{1}{T}\sum_{t=1}^T\mathbf{b}_t\mathbf{x}_t^\top-\mathbb{E}(\mathbf{b}_t\mathbf{x}_t^\top)-\mathrm{vec}\big(\bar{\mathbf{B}}\big)\frac{1}{T}\sum_{t=1}^T\mathbf{x}_t^\top\right\|_{\max}\leq c_T+\left\|\mathrm{vec}\big(\bar{\mathbf{B}}\big)\frac{1}{T}\sum_{t=1}^T\mathbf{x}_t^\top\right\|_{\max}\\
&\leq c_T+\Big(\left\|\mathrm{vec}\big(\bar{\mathbf{B}}\big)-\mathbb{E}[\mathbf{B}_t]\right\|_{\max}+\left\|\mathbb{E}[\mathbf{B}_t]\right\|_{\max}\Big)\left\|\frac{1}{T}\sum_{t=1}^T\mathbf{x}_t^\top\right\|_{\max}\leq c_T+c_T(c_T+\mu_{b,\max}).
\end{aligned}
\tag{S4.6}
$$

Similarly for I_2, we have on the set $\mathcal{M}$ that

$$
\begin{aligned}
&\left\|I_2\right\|_1\leq\frac{r^{1/2}}{d^2u^2}\cdot\left\|\mathbf{V}_{\mathbf{I}_d,r}^\top\mathbf{U}^\top\mathbf{V}_{\mathbf{I}_d,v}\right\|_1\cdot\left\|T^{-1}(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu\right\|_1\\
&=O\Big(\frac{1}{d}\Big)\left\|T^{-1}(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu\right\|_1=O\Big(\frac{1}{d}\Big)\left\|\frac{1}{T}\sum_{t=1}^T(\mathbf{B}_t^\top-\bar{\mathbf{B}}^\top)\boldsymbol{\epsilon}_t\right\|_1\\
&=O_p\Big(\frac{1}{d}\Big)\Big(c_Td^{\frac{1}{2}+\frac{1}{2w}}+c_Td^{1/2}\log^{1/2}(T\vee d)S_\epsilon(\mu_{b,\max}+c_T)\Big)=O_p\big(c_Td^{-\frac{1}{2}+\frac{1}{2w}}\big),
\end{aligned}
\tag{S4.7}
$$

where the first equality used $\mathcal{A}_1,\mathcal{A}_4,\mathcal{A}_6$ in $\mathcal{M}$, the third used $\mathcal{A}_3,\mathcal{A}_7$ in $\mathcal{M}$, and the last used Assumption (R10).

For I_3, first recall that

$$
\boldsymbol{\Pi}_t^*=(\mathbf{I}_d-\mathbf{W}_t^*)^{-1}=\Big(\mathbf{I}_d-\sum_{j=1}^p\sum_{k=0}^{l_j}\phi_{j,k}^*z_{j,k,t}\mathbf{W}_j\Big)^{-1},
\tag{S4.8}
$$

and hence from Assumption (M2) (resp. (M2')) we have $\|\boldsymbol{\Pi}_t^*\|_\infty\leq 1/(1-\eta)=O(1)$ (resp.

$\|\mathbf{\Pi}_t^*\|_\infty = O_P(1)$) using that $(\mathbf{I}_d - \mathbf{W}_t^*)$ is strictly diagonally dominant. Then by (S4.4), we have on the set $\mathcal{M}$ that

$$\begin{aligned}
&\big\|I_3\big\|_1 \leq \frac{r^{1/2}}{d^2u^2}\cdot\big\|\mathbf{V}_{\mathbf{I}_d,r}^\top\mathbf{U}^\top\mathbf{V}_{\mathbf{I}_d,v}\big\|_1\cdot\Big\|T^{-1}(\mathbf{B}^\nu)^\top\sum_{j=1}^p\sum_{k=0}^{l_j}(\phi_{j,k}^*-\widetilde{\phi}_{j,k})\mathbf{W}_j^\otimes\mathbf{D}_{z_{j,k}}\mathbf{\Pi}^{*\otimes}\mathbf{X}\Big\|_1\cdot\big\|\boldsymbol{\beta}^*\big\|_1\\
&= O\Big(\frac{1}{d}\Big)\Big\|\frac{1}{T}\sum_{j=1}^p\sum_{k=0}^{l_j}(\phi_{j,k}^*-\widetilde{\phi}_{j,k})(\mathbf{B}^\nu)^\top\mathbf{W}_j^\otimes\mathbf{D}_{z_{j,k}}\mathbf{\Pi}^{*\otimes}\mathbf{X}\Big\|_1\\
&= O\Big(\frac{1}{d}\Big)\Big\|\frac{1}{T}\sum_{j=1}^p\sum_{k=0}^{l_j}(\phi_{j,k}^*-\widetilde{\phi}_{j,k})\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t^\top-\bar{\mathbf{B}}^\top)\mathbf{W}_j\mathbf{\Pi}_t^*\mathbf{X}_t\Big\|_1\\
&= O\Big(\frac{1}{d}\Big)\Big\{\max_{q\in[r]}\max_{s\in[v]}\Big|\frac{1}{T}\sum_{j=1}^p\sum_{k=0}^{l_j}(\phi_{j,k}^*-\widetilde{\phi}_{j,k})\sum_{t=1}^T\sum_{i=1}^d z_{j,k,t}\mathbf{W}_{j,\cdot i}^\top(\mathbf{B}_{t,\cdot s}-\bar{\mathbf{B}}_{\cdot s})\mathbf{X}_{t,\cdot q}^\top\mathbf{\Pi}_{t,i\cdot}^*\Big|\Big\}\\
&= O\Big(\frac{1}{d}\Big)\Big(\big\|\boldsymbol{\phi}^*-\widetilde{\boldsymbol{\phi}}\big\|_1\\
&\qquad\cdot\max_{q\in[r],s\in[v],j\in[p]}\max_{k\in[l_j]\cup\{0\}}\max_{m,n\in[d]}\Big|\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(B_{t,ms}-\bar{B}_{ms})X_{t,nq}\Big|\sum_{i=1}^d\|\mathbf{W}_{j,\cdot i}\|_1\|\mathbf{\Pi}_{t,i\cdot}^*\|_1\Big)\\
&= O_p\Big(\big\|\boldsymbol{\phi}^*-\widetilde{\boldsymbol{\phi}}\big\|_1\big[c_T+1+c_T(c_T+\mu_{b,\max})\big]\Big) = O_p\Big(\big\|\boldsymbol{\phi}^*-\widetilde{\boldsymbol{\phi}}\big\|_1\Big),
\end{aligned} \tag{S4.9}$$

where the first equality used $\mathcal{A}_1, \mathcal{A}_4, \mathcal{A}_6$ in $\mathcal{M}$, the second last used $\mathcal{A}_4, \mathcal{A}_8, \mathcal{A}_9$ in $\mathcal{M}$ and Assumptions (R1) and (R3). Similarly, for I_4 on the set $\mathcal{M}$,

$$\begin{aligned}
&\big\|I_4\big\|_1 \leq \frac{r^{1/2}}{d^2u^2}\cdot\big\|\mathbf{V}_{\mathbf{I}_d,r}^\top\mathbf{U}^\top\mathbf{V}_{\mathbf{I}_d,v}\big\|_1\cdot\big\|T^{-1}(\mathbf{B}^\nu)^\top\sum_{j=1}^p\sum_{k=0}^{l_j}(\phi_{j,k}^*-\widetilde{\phi}_{j,k})\mathbf{W}_j^\otimes\mathbf{D}_{z_{j,k}}\mathbf{\Pi}^{*\otimes}\boldsymbol{\epsilon}^\nu\big\|_1\\
&= O\Big(\frac{1}{d}\Big)\Big\{\max_{s\in[v]}\Big|\frac{1}{T}\sum_{j=1}^p\sum_{k=0}^{l_j}(\phi_{j,k}^*-\widetilde{\phi}_{j,k})\sum_{t=1}^T\sum_{i=1}^d z_{j,k,t}\mathbf{W}_{j,\cdot i}^\top(\mathbf{B}_{t,\cdot s}-\bar{\mathbf{B}}_{\cdot s})\boldsymbol{\epsilon}_t^\top\mathbf{\Pi}_{t,i\cdot}^*\Big|\Big\}\\
&= O\Big(\frac{1}{d}\Big)\Big\{\big\|\boldsymbol{\phi}^*-\widetilde{\boldsymbol{\phi}}\big\|_1\\
&\qquad\cdot\max_{s\in[v]}\max_{j\in[p]}\max_{k\in[l_j]\cup\{0\}}\max_{m,q\in[d]}\Big|\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(B_{t,ms}-\bar{B}_{ms})\epsilon_{t,q}\Big|\sum_{i=1}^d\|\mathbf{W}_{j,\cdot i}\|_1\|\mathbf{\Pi}_{t,i\cdot}^*\|_1\Big\}\\
&= O_p\Big(\big\|\boldsymbol{\phi}^*-\widetilde{\boldsymbol{\phi}}\big\|_1\big[c_T+c_T(c_T+\mu_{b,\max})\big]\Big) = O_p\Big(c_T\big\|\boldsymbol{\phi}^*-\widetilde{\boldsymbol{\phi}}\big\|_1\Big),
\end{aligned} \tag{S4.10}$$

where the first equality used $\mathcal{A}_1, \mathcal{A}_4, \mathcal{A}_6$ in $\mathcal{M}$, the second last used $\mathcal{A}_4, \mathcal{A}_{10}, \mathcal{A}_{11}$ in $\mathcal{M}$ and

Assumptions (R1) and (R3). For I_5 we also have on the set $\mathcal{M}$,

$$\begin{aligned}
&\big\|I_5\big\|_1 \le \frac{r^{1/2}}{d^2u^2}\cdot\big\|\mathbf{V}_{\mathbf{I}_d,r}^\top\mathbf{U}^\top\mathbf{V}_{\mathbf{I}_d,v}\big\|_1\cdot\big\|T^{-1}(\mathbf{B}^\nu)^\top\sum_{j=1}^p\sum_{k=0}^{l_j}(\phi_{j,k}^*-\widetilde{\phi}_{j,k})\mathbf{W}_j^\otimes\mathbf{D}_{z_{j,k}}\mathbf{\Pi}^{*\otimes}(\mathbf{1}_T\otimes\boldsymbol{\mu}^*)\big\|_1\\
&=O\Big(\frac{1}{d}\Big)\Big\{\max_{s\in[v]}\Big|\frac{1}{T}\sum_{j=1}^p\sum_{k=0}^{l_j}(\phi_{j,k}^*-\widetilde{\phi}_{j,k})\sum_{t=1}^T\sum_{i=1}^d z_{j,k,t}\mathbf{W}_{j,\cdot i}^\top(\mathbf{B}_{t,\cdot s}-\bar{\mathbf{B}}_{\cdot s})\boldsymbol{\mu}^{*\top}\mathbf{\Pi}_{t,i\cdot}^*\Big|\Big\}\\
&=O\Big(\frac{1}{d}\Big)\Big\{\big\|\boldsymbol{\phi}^*-\widetilde{\boldsymbol{\phi}}\big\|_1\\
&\qquad\cdot\max_{s\in[v]}\max_{j\in[p]}\max_{k\in[l_j]\cup\{0\}}\max_{m\in[d]}\Big|\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(B_{t,ms}-\bar{B}_{ms})\Big|\|\boldsymbol{\mu}^*\|_{\max}\sum_{i=1}^d\|\mathbf{W}_{j,\cdot i}\|_1\|\mathbf{\Pi}_{t,i\cdot}^*\|_1\Big\}\\
&=O_p\Big(\big\|\boldsymbol{\phi}^*-\widetilde{\boldsymbol{\phi}}\big\|_1\big[c_T+(z_{\max}c_T\vee c_T)\big]\Big)=O_p\Big(c_T\big\|\boldsymbol{\phi}^*-\widetilde{\boldsymbol{\phi}}\big\|_1\Big),
\end{aligned}\tag{S4.11}$$

where the first equality used $\mathcal{A}_1,\mathcal{A}_4,\mathcal{A}_6$ in $\mathcal{M}$, the second last used $\mathcal{A}_4,\mathcal{A}_{12},\mathcal{A}_{13}$ in $\mathcal{M}$ and Assumptions (R1), (R3), and $\mathbb{E}(z_{j,k,t}\mathbf{B}_t)=\mathbf{0}$ from (M2') if (M2) is not satisfied.

From (S4.5), (S4.7), (S4.9) and (S4.10), combining with Lemma S2, we have

$$\big\|\boldsymbol{\beta}(\widetilde{\boldsymbol{\phi}})-\boldsymbol{\beta}^*\big\|_1\le\sum_{j=1}^5\big\|I_j\big\|_1=O_p\Big(\big\|\widetilde{\boldsymbol{\phi}}-\boldsymbol{\phi}^*\big\|_1+c_Td^{-\frac{1}{2}+\frac{1}{2w}}\Big).\tag{S4.12}$$

For the remaining of the proof of 1, we work on the rate for $\big\|\widetilde{\boldsymbol{\phi}}-\boldsymbol{\phi}^*\big\|_1$. From (2.14), we have

$$\begin{aligned}
\widetilde{\boldsymbol{\phi}}&=\big[(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)\big]^{-1}(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top(\mathbf{B}^\top\mathbf{y}-\mathbf{\Xi}\mathbf{y}^\nu)\\
&=\big[(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)\big]^{-1}(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top(\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}^*}\mathrm{vec}\big(\mathbf{I}_d\big)+\mathbf{B}^\top\boldsymbol{\epsilon}-\mathbf{\Xi}\mathbf{y}^\nu)\\
&\quad+\big[(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)\big]^{-1}(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top(\mathbf{\Xi}\mathbf{Y}_W)\boldsymbol{\phi}^*+\boldsymbol{\phi}^*\\
&=\boldsymbol{\phi}^*+\big[(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)\big]^{-1}(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top\mathbf{B}^\top\boldsymbol{\epsilon}\\
&\quad-T^{-1/2}d^{-a/2}\cdot\big[(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)\big]^{-1}(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)^\top\\
&\quad\cdot\mathrm{vec}\Big\{\sum_{t=1}^T(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}(\boldsymbol{\epsilon}^\nu)^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X})^{-1}\mathbf{X}_t^\top\Big\},
\end{aligned}$$

where the second equality used (2.11), and the third used the fact that $\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}(\boldsymbol{\phi})}\mathrm{vec}\big(\mathbf{I}_d\big)=\mathbf{\Xi}\mathbf{y}^\nu-\mathbf{\Xi}\mathbf{Y}_W\boldsymbol{\phi}$ from (S1.3) and $\boldsymbol{\beta}(\boldsymbol{\phi}^*)=\boldsymbol{\beta}^*+\big(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\big)^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu$. Thus, we may decompose

$$\widetilde{\boldsymbol{\phi}}-\boldsymbol{\phi}^*=D_1-D_2,\text{where}$$

$$D_1 = \left[(\mathbf{B}^\top \mathbf{V} - \boldsymbol{\Xi}\mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \boldsymbol{\Xi}\mathbf{Y}_W)\right]^{-1} (\mathbf{B}^\top \mathbf{V} - \boldsymbol{\Xi}\mathbf{Y}_W)^\top \mathbf{B}^\top \boldsymbol{\epsilon},$$

$$D_2 = T^{-1/2} d^{-a/2} \cdot \left[(\mathbf{B}^\top \mathbf{V} - \boldsymbol{\Xi}\mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \boldsymbol{\Xi}\mathbf{Y}_W)\right]^{-1} (\mathbf{B}^\top \mathbf{V} - \boldsymbol{\Xi}\mathbf{Y}_W)^\top \cdot \left\{\sum_{t=1}^T \left(\mathbf{I}_d \otimes (\mathbf{B}_t - \bar{\mathbf{B}})\boldsymbol{\gamma}\right)\mathbf{X}_t\right\} (\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu.$$

To bound the above, recall first the following definitions (from the statement of Theorem 2) for $j \in [p], k \in [l_j] \cup \{0\}$,

$$\mathbf{U}_{\mathbf{x},j,k} := \frac{1}{T}\sum_{t=1}^T z_{j,k,t} \mathrm{vec}\left(\mathbf{B}_t - \bar{\mathbf{B}}\right)\mathbf{x}_t^\top, \mathbf{U}_{\boldsymbol{\mu},j,k} := \frac{1}{T}\sum_{t=1}^T z_{j,k,t} \mathrm{vec}\left(\mathbf{B}_t - \bar{\mathbf{B}}\right)\boldsymbol{\mu}^{*\top},$$

$$\mathbf{U}_{\boldsymbol{\epsilon},j,k} := \frac{1}{T}\sum_{t=1}^T z_{j,k,t} \mathrm{vec}\left(\mathbf{B}_t - \bar{\mathbf{B}}\right)\boldsymbol{\epsilon}_t^\top.$$

Consider $\boldsymbol{\Xi}\mathbf{Y}_W$, from its definition we have,

$$\begin{aligned}
\boldsymbol{\Xi}\mathbf{Y}_W &= T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}})\boldsymbol{\gamma}\right) \left[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}\right]^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \\
&\quad \cdot (\mathbf{W}_1^\otimes \mathbf{y}_{1,0}^\nu, \ldots, \mathbf{W}_1^\otimes \mathbf{y}_{1,l_1}^\nu, \ldots, \mathbf{W}_p^\otimes \mathbf{y}_{p,0}^\nu, \ldots, \mathbf{W}_p^\otimes \mathbf{y}_{p,l_p}^\nu) \\
&= T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}})\boldsymbol{\gamma}\right) \left[\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}\right]^{-1} \mathbf{X}^\top \mathbf{B}^\nu \\
&\quad \cdot \left\{\sum_{t=1}^T z_{1,0,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_1 \mathbf{y}_t, \cdots, \sum_{t=1}^T z_{p,l_p,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_p \mathbf{y}_t\right\}
\end{aligned}$$

From (S4.8), we have $\mathbf{y}_t = \boldsymbol{\Pi}_t^* \boldsymbol{\mu}^* + \boldsymbol{\Pi}_t^* \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\Pi}_t^* \boldsymbol{\epsilon}_t$. It hence holds for any $j \in [p], k \in [l_j] \cup \{0\}$ by Lemma S3 that

$$\begin{aligned}
&\frac{1}{T}\sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_j \mathbf{y}_t \\
&= \frac{1}{T}\sum_{t=1}^T z_{j,k,t} \mathbf{V}_{\mathbf{W}_j^\top, v}^\top \left(\mathbf{I}_d \otimes \mathrm{vec}\left(\mathbf{B}_t - \bar{\mathbf{B}}\right)\right) (\boldsymbol{\Pi}_t^* \boldsymbol{\mu}^* + \boldsymbol{\Pi}_t^* \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\Pi}_t^* \boldsymbol{\epsilon}_t) \\
&= \mathbf{V}_{\mathbf{W}_j^\top, v}^\top (\mathbf{I}_d \otimes \mathbf{U}_{\mathbf{x},j,k}) \mathbf{V}_{\boldsymbol{\Pi}_t^*, r} \boldsymbol{\beta}^* + \mathbf{V}_{\mathbf{W}_j^\top, v}^\top (\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\mu},j,k}) \mathrm{vec}\left(\boldsymbol{\Pi}_t^{*\top}\right) + \mathbf{V}_{\mathbf{W}_j^\top, v}^\top (\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\epsilon},j,k}) \mathrm{vec}\left(\boldsymbol{\Pi}_t^{*\top}\right).
\end{aligned} \quad \text{(S4.13)}$$

Consider also $\mathbf{B}^\top\mathbf{V}$, we have

$$
\begin{aligned}
\mathbf{B}^\top\mathbf{V} &= T^{-1/2}d^{-a/2}\Big(\mathbf{I}_d \otimes \{(\mathbf{B}_1-\bar{\mathbf{B}},\ldots,\mathbf{B}_T-\bar{\mathbf{B}})(\mathbf{I}_T\otimes\boldsymbol{\gamma})\}\Big) \\
&\cdot\Big\{\Big[\mathbf{I}_d\otimes(z_{1,0,1}\mathbf{y}_1,\ldots,z_{1,0,T}\mathbf{y}_T)^\top\Big]\mathrm{vec}\big(\mathbf{W}_1^\top\big),\ldots,\Big[\mathbf{I}_d\otimes(z_{1,l_1,1}\mathbf{y}_1,\ldots,z_{1,l_1,T}\mathbf{y}_T)^\top\Big]\mathrm{vec}\big(\mathbf{W}_1^\top\big), \\
&\ldots,\Big[\mathbf{I}_d\otimes(z_{p,0,1}\mathbf{y}_1,\ldots,z_{p,0,T}\mathbf{y}_T)^\top\Big]\mathrm{vec}\big(\mathbf{W}_p^\top\big),\ldots,\Big[\mathbf{I}_d\otimes(z_{p,l_p,1}\mathbf{y}_1,\ldots,z_{p,l_p,T}\mathbf{y}_T)^\top\Big]\mathrm{vec}\big(\mathbf{W}_p^\top\big)\Big\} \\
&= T^{-1/2}d^{-a/2}\Big\{\Big[\mathbf{I}_d\otimes(\mathbf{B}_1-\bar{\mathbf{B}},\ldots,\mathbf{B}_T-\bar{\mathbf{B}})(\mathbf{I}_T\otimes\boldsymbol{\gamma})(z_{1,0,1}\mathbf{y}_1,\ldots,z_{1,0,T}\mathbf{y}_T)^\top\Big]\mathrm{vec}\big(\mathbf{W}_1^\top\big), \\
&\ldots,\Big[\mathbf{I}_d\otimes(\mathbf{B}_1-\bar{\mathbf{B}},\ldots,\mathbf{B}_T-\bar{\mathbf{B}})(\mathbf{I}_T\otimes\boldsymbol{\gamma})(z_{p,l_p,1}\mathbf{y}_1,\ldots,z_{p,l_p,T}\mathbf{y}_T)^\top\Big]\mathrm{vec}\big(\mathbf{W}_p^\top\big)\Big\} \\
&= T^{-1/2}d^{-a/2}\Big\{\Big[\mathbf{I}_d\otimes\sum_{t=1}^T z_{1,0,t}(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\mathbf{y}_t^\top\Big]\mathrm{vec}\big(\mathbf{W}_1^\top\big), \\
&\qquad\ldots,\Big[\mathbf{I}_d\otimes\sum_{t=1}^T z_{p,l_p,t}(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\mathbf{y}_t^\top\Big]\mathrm{vec}\big(\mathbf{W}_p^\top\big)\Big\}.
\end{aligned}
$$

Similar to $\boldsymbol{\Xi}\mathbf{Y}_W$, for $j\in[p], k\in[l_p]\cup\{0\}$ we have

$$
\begin{aligned}
&\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\mathbf{y}_t^\top = \frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}(\boldsymbol{\mu}^{*\top}\boldsymbol{\Pi}_t^{*\top}+\boldsymbol{\beta}^{*\top}\mathbf{X}_t^\top\boldsymbol{\Pi}_t^{*\top}+\boldsymbol{\epsilon}_t^\top\boldsymbol{\Pi}_t^{*\top}) \\
&= \frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathrm{vec}\big(\mathbf{B}_t-\bar{\mathbf{B}}\big)\boldsymbol{\mu}^{*\top}\boldsymbol{\Pi}_t^{*\top} + \frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathrm{vec}\big(\mathbf{B}_t-\bar{\mathbf{B}}\big)\mathbf{x}_t^\top(\boldsymbol{\beta}^*\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top} \\
&+ \frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathrm{vec}\big(\mathbf{B}_t-\bar{\mathbf{B}}\big)\boldsymbol{\epsilon}_t^\top\boldsymbol{\Pi}_t^{*\top} \\
&= (\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbf{U}_{\boldsymbol{\mu},j,k}\boldsymbol{\Pi}_t^{*\top} + (\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbf{U}_{\mathbf{x},j,k}(\boldsymbol{\beta}^*\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top} + (\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbf{U}_{\boldsymbol{\epsilon},j,k}\boldsymbol{\Pi}_t^{*\top}.
\end{aligned}
\tag{S4.14}
$$

With (S4.13) and (S4.14), recall the following in the statement of Theorem 2,

$$
\begin{aligned}
\mathbf{H}_{10} = \Big\{&\Big(\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbb{E}\big(\mathbf{U}_{\mathbf{x},1,0}(\boldsymbol{\beta}^*\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}\big)\Big)\mathrm{vec}(\mathbf{W}_1^\top) \\
&+\Big(\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbb{E}\big(\mathbf{U}_{\boldsymbol{\mu},1,0}\boldsymbol{\Pi}_t^{*\top}\big)\Big)\mathrm{vec}(\mathbf{W}_1^\top), \\
\ldots,&\Big(\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbb{E}\big(\mathbf{U}_{\mathbf{x},p,l_p}(\boldsymbol{\beta}^*\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}\big)\Big)\mathrm{vec}(\mathbf{W}_p^\top) \\
&+\Big(\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbb{E}\big(\mathbf{U}_{\boldsymbol{\mu},p,l_p}\boldsymbol{\Pi}_t^{*\top}\big)\Big)\mathrm{vec}(\mathbf{W}_p^\top)\Big\},
\end{aligned}
$$

$$
\begin{aligned}
\mathbf{H}_{20}=&\mathbb{E}(\mathbf{X}_t\otimes\mathbf{B}_t\boldsymbol{\gamma})\big[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\big]^{-1}\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\\
&\cdot\Big\{\mathbf{V}_{\mathbf{W}_1^\top,v}^\top\mathbb{E}\big[(\mathbf{I}_d\otimes\mathbf{U}_{\mathbf{x},1,0})\mathbf{V}_{\boldsymbol{\Pi}_t^*,r}\big]\boldsymbol{\beta}^*+\mathbf{V}_{\mathbf{W}_1^\top,v}^\top\mathbb{E}\big[(\mathbf{I}_d\otimes\mathbf{U}_{\boldsymbol{\mu},1,0})\mathrm{vec}(\boldsymbol{\Pi}_t^{*\top})\big],\\
&...,\mathbf{V}_{\mathbf{W}_p^\top,v}^\top\mathbb{E}\big[(\mathbf{I}_d\otimes\mathbf{U}_{\mathbf{x},p,l_p})\mathbf{V}_{\boldsymbol{\Pi}_t^*,r}\big]\boldsymbol{\beta}^*+\mathbf{V}_{\mathbf{W}_p^\top,v}^\top\mathbb{E}\big[(\mathbf{I}_d\otimes\mathbf{U}_{\boldsymbol{\mu},p,l_p})\mathrm{vec}(\boldsymbol{\Pi}_t^{*\top})\big]\Big\},
\end{aligned}
$$

which are essentially $T^{-1/2}d^{a/2}\mathbf{B}^\top\mathbf{V}$ and $T^{-1/2}d^{a/2}\boldsymbol{\Xi}\mathbf{Y}_W$ at the population level, respectively.

For the rest of the proof of Theorem 1, we find the rate of D_2, followed by constructing the asymptotic normality of the dominating term in the expansion of D_1. For D_2, we further decompose $D_2=F_1+F_2-F_3$ where

$$
\begin{aligned}
F_1=&[(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})]^{-1}\Big\{(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})\\
&-T^{-1}d^a(\mathbf{B}^\top\mathbf{V}-\boldsymbol{\Xi}\mathbf{Y}_W)^\top(\mathbf{B}^\top\mathbf{V}-\boldsymbol{\Xi}\mathbf{Y}_W)\Big\}D_2,\\
F_2=&[(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})]^{-1}\Big[(T^{-1/2}d^{a/2}\mathbf{B}^\top\mathbf{V}-\mathbf{H}_{10})-(T^{-1/2}d^{a/2}\boldsymbol{\Xi}\mathbf{Y}_W-\mathbf{H}_{20})\Big]^\top\\
&\cdot\Big\{\frac{1}{T}\sum_{t=1}^T\Big(\mathbf{I}_d\otimes(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\Big)\mathbf{X}_t\Big\}(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu,\\
F_3=&[(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})]^{-1}(\mathbf{H}_{20}-\mathbf{H}_{10})^\top\\
&\cdot\Big\{\frac{1}{T}\sum_{t=1}^T\Big(\mathbf{I}_d\otimes(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\Big)\mathbf{X}_t\Big\}(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu.
\end{aligned}
$$

To bound the L1 norm of F_1 to F_3, first observe that by Assumptions (R1) and (R4) we have

$$
\begin{aligned}
\sigma_L^2(\mathbf{H}_{10})\geq&\,\sigma_L^2\Bigg(\Big\{\Big(\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbb{E}\big(\mathbf{U}_{\mathbf{x},1,0}(\boldsymbol{\beta}^*\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}\big)\Big)\mathrm{vec}\big(\mathbf{W}_1^\top\big),\\
&...,\Big(\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbb{E}\big(\mathbf{U}_{\mathbf{x},p,l_p}(\boldsymbol{\beta}^*\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}\big)\Big)\mathrm{vec}\big(\mathbf{W}_p^\top\big)\Big\}\Bigg)\\
\geq&\,\sigma_L^2(\mathbf{D}_W)\cdot\sigma_{d^2}^2\Bigg(\Big\{\Big(\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbb{E}\big(\mathbf{U}_{\mathbf{x},1,0}(\boldsymbol{\beta}^*\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}\big)\Big),\\
&...,\Big(\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbb{E}\big(\mathbf{U}_{\mathbf{x},p,l_p}(\boldsymbol{\beta}^*\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}\big)\Big)\Big\}\Bigg)\geq Cd\cdot d^a=Cd^{1+a},
\end{aligned}
$$

where $C>0$ is a generic constant. Similarly, by Assumptions (R1), (R3), (R5) and (R6),

$$
\sigma_L(\mathbf{H}_{20})\geq\sigma_r\Big(\mathbb{E}(\mathbf{X}_t\otimes\mathbf{B}_t\boldsymbol{\gamma})\Big)\sigma_r\Big(\big[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\big]^{-1}\Big)\sigma_r\Big(\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\Big)
$$

$$
\begin{aligned}
&\cdot\sigma_{\min}\Big(\Big\{\mathbf{V}_{\mathbf{W}_1^\top,r}^\top\mathbb{E}\big[(\mathbf{I}_d\otimes\mathbf{U}_{\mathbf{x},1,0})\mathbf{V}_{\mathbf{\Pi}_t^*,r}\big]\boldsymbol{\beta}^*,\ldots,\mathbf{V}_{\mathbf{W}_p^\top,r}^\top\mathbb{E}\big[(\mathbf{I}_d\otimes\mathbf{U}_{\mathbf{x},p,l_p})\mathbf{V}_{\mathbf{\Pi}_t^*,r}\big]\boldsymbol{\beta}^*\Big\}\Big)\\
&\geq\frac{Cd^{1+a}\cdot d}{\lambda_{\max}[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)]}\cdot\sigma_v[\mathbb{E}(\ddot{\mathbf{G}})]\cdot\sigma_L(\mathbf{D}_W)\\
&\geq\frac{Cd^{1+a}\cdot d\cdot d^{1/2}}{\lambda_{\max}[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)]}\geq Cd^{1/2+a},
\end{aligned}
$$

with some arbitrary constant $C>0$. Notice $\mathbf{H}_{20}$ has the smallest singular value of order larger then that for $\mathbf{H}_{10}$, so $\sigma_L^2(\mathbf{H}_{20}-\mathbf{H}_{10})\geq Cd^{1+a}$ for some $C>0$. Thus,

$$
\Big\|\big[(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})\big]^{-1}\Big\|_1\leq\frac{L^{1/2}}{\lambda_{\min}[(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})]}\leq\frac{L^{1/2}}{Cd^{1+a}}. \tag{S4.15}
$$

Consider F_1 first and we hence have on $\mathcal{M}$,

$$
\begin{aligned}
\|F_1\|_1\leq\frac{L^{1/2}\cdot L}{Cd^{1+a}}\bigg\{&\Big\|(\mathbf{H}_{20}-T^{-1/2}d^{a/2}\mathbf{\Xi}\mathbf{Y}_W)+(T^{-1/2}d^{a/2}\mathbf{B}^\top\mathbf{V}-\mathbf{H}_{10})\Big\|_{\max}\cdot\big\|\mathbf{H}_{20}-\mathbf{H}_{10}\big\|_1\\
&+\Big\|T^{-1/2}d^{a/2}(\mathbf{B}^\top\mathbf{V}-\mathbf{\Xi}\mathbf{Y}_W)\Big\|_{\max}\\
&\cdot\Big\|(\mathbf{H}_{20}-T^{-1/2}d^{a/2}\mathbf{\Xi}\mathbf{Y}_W)+(T^{-1/2}d^{a/2}\mathbf{B}^\top\mathbf{V}-\mathbf{H}_{10})\Big\|_1\bigg\}\|D_2\|_1\\
=O\Big\{&\frac{L^{3/2}}{d^{1+a}}[c_T\cdot d^2+1\cdot(c_Td^2+c_Td^2)]\Big\}\|D_2\|_1=O\Big(c_TL^{3/2}d^{1-a}\|D_2\|_1\Big),
\end{aligned}
\tag{S4.16}
$$

where the last line used the following rates to be shown later,

$$
\big\|\mathbf{H}_{20}-T^{-1/2}d^{a/2}\mathbf{\Xi}\mathbf{Y}_W\big\|_{\max}=O_P(c_T), \tag{S4.17}
$$

$$
\big\|\mathbf{H}_{10}-T^{-1/2}d^{a/2}\mathbf{B}^\top\mathbf{V}\big\|_{\max}=O_P(c_T). \tag{S4.18}
$$

For neat presentation, we define the following terms whose norms will be bounded and involved in (S4.17) and later,

$$
\begin{aligned}
&\mathbf{A}_1:=\frac{1}{T}\sum_{t=1}^T\mathbf{X}_t\otimes(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}, &&\mathbf{A}_1^0:=\mathbb{E}(\mathbf{X}_t\otimes\mathbf{B}_t\boldsymbol{\gamma}),\\
&\mathbf{A}_2:=\Big(\frac{1}{T}\mathbf{X}^\top\mathbf{B}^\nu\frac{1}{T}(\mathbf{B}^\nu)^\top\mathbf{X}\Big)^{-1}, &&\mathbf{A}_2^0:=\big[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\big]^{-1},\\
&\mathbf{A}_3:=\frac{1}{T}\mathbf{X}^\top\mathbf{B}^\nu, &&\mathbf{A}_3^0:=\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t),\\
&\mathbf{A}_{4,j,k}:=\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})^\top\mathbf{W}_j(\mathbf{\Pi}_t^*\boldsymbol{\mu}^*+\mathbf{\Pi}_t^*\mathbf{X}_t\boldsymbol{\beta}^*),
\end{aligned}
$$

$$\mathbf{A}_{4,j,k}^0 := \mathbb{E}(\mathbf{A}_{4,j,k}), \qquad \mathbf{A}_{5,j,k} := \frac{1}{T}\sum_{t=1}^{T} z_{j,k,t}(\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_j \mathbf{\Pi}_t^* \boldsymbol{\epsilon}_t.$$

On $\mathcal{M}$, we immediately have from Lemma S2 (using $\mathcal{A}_1, \mathcal{A}_4, \mathcal{A}_6$),

$$\|\mathbf{A}_1 - \mathbf{A}_1^0\|_{\max} = O\Big(c_T + c_T(c_T + \mu_{b,\max})\Big) = O(c_T), \tag{S4.19}$$

which also gives $\|\mathbf{A}_1\|_{\max} \le \|\mathbf{A}_1^0\|_{\max} + \|\mathbf{A}_1 - \mathbf{A}_1^0\|_{\max} = O(1 + c_T) = O(1)$. Hence with Assumptions (R5) and (R10), we also have on $\mathcal{M}$ that

$$\|\mathbf{A}_1\|_1 \le \|\mathbf{A}_1^0\|_1 + \|\mathbf{A}_1 - \mathbf{A}_1^0\|_1 = O(d^{1+a} + c_T d^2) = O(d^{1+a}). \tag{S4.20}$$

Similarly, with Lemma S2 (using $\mathcal{A}_1$) and Assumption (R3) we have on $\mathcal{M}$,

$$\|\mathbf{A}_3 - \mathbf{A}_3^0\|_1 = O(c_T d), \|\mathbf{A}_3^0\|_1 = O(d), \|\mathbf{A}_3\|_1 \le \|\mathbf{A}_3 - \mathbf{A}_3^0\|_1 + \|\mathbf{A}_3^0\|_1 = O(d). \tag{S4.21}$$

Rewrite $\mathbf{A}_2^0 = (\mathbf{A}_3^0 \mathbf{A}_3^{0\top})^{-1}$ and $\mathbf{A}_2 = (\mathbf{A}_3 \mathbf{A}_3^\top)^{-1}$, by Assumption (R3),

$$\|\mathbf{A}_2^0\|_1 \le \frac{r^{1/2}}{\lambda_{\min}(\mathbf{A}_3^0 \mathbf{A}_3^{0\top})} = O(d^{-2}). \tag{S4.22}$$

Moreover, from (S4.21) we have on $\mathcal{M}$

$$\begin{aligned}
\|(\mathbf{A}_2^0)^{-1} - \mathbf{A}_2^{-1}\|_1 &\le \|\mathbf{A}_3^0 \mathbf{A}_3^{0\top} - \mathbf{A}_3 \mathbf{A}_3^\top\|_1 \le \|\mathbf{A}_3^0 - \mathbf{A}_3\|_1 \|\mathbf{A}_3^{0\top}\|_1 + \|\mathbf{A}_3\|_1 \|\mathbf{A}_3^{0\top} - \mathbf{A}_3^\top\|_1 \\
&= O(c_T d \cdot d + d \cdot c_T d) = O(c_T d^2).
\end{aligned}$$

Thus, rewriting $\mathbf{A}_2 - \mathbf{A}_2^0 = (\mathbf{A}_2 - \mathbf{A}_2^0)[(\mathbf{A}_2^0)^{-1} - \mathbf{A}_2^{-1}]\mathbf{A}_2^0 + \mathbf{A}_2^0[(\mathbf{A}_2^0)^{-1} - \mathbf{A}_2^{-1}]\mathbf{A}_2^0$ and we have on $\mathcal{M}$,

$$\|\mathbf{A}_2 - \mathbf{A}_2^0\|_1 = o\Big(\|\mathbf{A}_2 - \mathbf{A}_2^0\|_1\Big) + O(c_T d^2 \cdot d^{-4}) = O(c_T d^{-2}). \tag{S4.23}$$

Consider now $\mathbf{A}_{4,j,k}^0$ for any $j \in [p], k \in [l_j] \cup \{0\}$. First, we have on $\mathcal{M}$,

$$\begin{aligned}
&\Big\|\mathbb{E}\Big(\frac{1}{T}\sum_{t=1}^{T} z_{j,k,t}(\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_j \mathbf{\Pi}_t^* \boldsymbol{\mu}^*\Big)\Big\|_1 \le \Big\|\mathbb{E}\Big(\frac{1}{T}\sum_{t=1}^{T} z_{j,k,t}(\mathbf{B}_t - \bar{\mathbf{B}})^\top\Big)\Big\|_{\max} \|\mathbf{W}_j\|_1 \|\mathbf{\Pi}_t^* \boldsymbol{\mu}^*\|_1 \\
&= O\Big(d \cdot \|\mathbf{\Pi}_t^*\|_\infty \|\boldsymbol{\mu}^*\|_{\max}\Big) = O(d),
\end{aligned}$$

where the last equality used Assumption (R1). Thus on $\mathcal{M}$,

$$
\begin{aligned}
&\big\|\mathbf{A}^0_{4,j,k}\big\|_1 \le \big\|\mathbf{V}^\top_{\mathbf{W}_j^\top,v}\big[\mathbf{I}_d\otimes\mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})\big]\mathbf{V}_{\mathbf{\Pi}_t^*,r}\boldsymbol{\beta}^*\big\|_1+\Big\|\mathbb{E}\Big(\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})^\top\mathbf{W}_j\mathbf{\Pi}_t^*\boldsymbol{\mu}^*\Big)\Big\|_1\\
&\le \big\|\mathbf{V}^\top_{\mathbf{W}_j^\top,v}\big[\mathbf{I}_d\otimes\mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})\big]\big\|_{\max}\big\|\mathbf{V}_{\mathbf{\Pi}_t^*,r}\big\|_1\big\|\boldsymbol{\beta}^*\big\|_1+O(d)\\
&\le \max_{i,n\in[d]}\max_{m\in[v]}\max_{q\in[r]}\mathbf{W}^\top_{j,\cdot i}\mathbb{E}\Big\{\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})_{\cdot m}X_{t,nq}\Big\}\big\|\mathbf{V}_{\mathbf{\Pi}_t^*,r}\big\|_1\big\|\boldsymbol{\beta}^*\big\|_1+O(d)=O(d),
\end{aligned}
\tag{S4.24}
$$

where the last equality used Assumption (R1). Furthermore, we have on $\mathcal{M}$ that

$$
\begin{aligned}
&\big\|\mathbf{A}_{4,j,k}-\mathbf{A}^0_{4,j,k}\big\|_1\\
&\le\Big\|\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})^\top-\mathbb{E}\Big\{\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})^\top\Big\}\Big\|_{\max}\big\|\mathbf{W}_j\big\|_1\big\|\mathbf{\Pi}_t^*\boldsymbol{\mu}^*\big\|_1\\
&\quad+\big\|\mathbf{V}^\top_{\mathbf{W}_j^\top,v}\big\{\mathbf{I}_d\otimes[\mathbf{U}_{\mathbf{x},j,k}-\mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})]\big\}\mathbf{V}_{\mathbf{\Pi}_t^*,r}\boldsymbol{\beta}^*\big\|_1\\
&=O\Big([c_T+1\cdot c_T+1\cdot c_T+1\cdot(c_T\vee 0)]d\Big)+O\Big(\big\|\mathbf{V}_{\mathbf{\Pi}_t^*,r}\big\|_1\big\|\boldsymbol{\beta}^*\big\|_1\Big)\\
&\quad\cdot\big\|\mathbf{W}_j^\top\big\|_1\max_{n,m\in[d]}\max_{s\in[v]}\max_{q\in[r]}\Big|\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})_{ms}X_{t,nq}-\mathbb{E}\Big(\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})_{ms}X_{t,nq}\Big)\Big|\\
&=O(c_Td),
\end{aligned}
\tag{S4.25}
$$

where the second last equality used Assumption (R1) and $\mathcal{A}_4,\mathcal{A}_{12},\mathcal{A}_{13}$, while the last used $\mathcal{A}_4$,$\mathcal{A}_8$,$\mathcal{A}_9$. In particular, we used the following as an immediate result of $\mathcal{A}_{13}$,

$$
\max_{m\in[p]}\max_{n\in[l_m]\cup\{0\}}\Big|\frac{1}{T}\sum_{t=1}^T z_{m,n,t}-\mathbb{E}\Big\{\frac{1}{T}\sum_{t=1}^T z_{m,n,t}\Big\}\Big|\le c_T\vee 0.
$$

Lastly for any $j\in[p],k\in[l_j]\cup\{0\}$, similar to (S4.10), we have on $\mathcal{M}$ that

$$
\begin{aligned}
\big\|\mathbf{A}_{5,j,k}\big\|_1&=\big\|\mathbf{V}^\top_{\mathbf{W}_j^\top,v}\big(\mathbf{I}_d\otimes\mathbf{U}_{\boldsymbol{\epsilon},j,k}\big)\mathbf{V}_{\mathbf{\Pi}_t^*,r}\big\|_1\le\big\|\mathbf{V}^\top_{\mathbf{W}_j^\top,v}\big(\mathbf{I}_d\otimes\mathbf{U}_{\boldsymbol{\epsilon},j,k}\big)\big\|_{\max}\big\|\mathbf{V}_{\mathbf{\Pi}_t^*,r}\big\|_1\\
&\le\big\|\mathbf{W}_j^\top\big\|_1\max_{m,n\in[d]}\max_{s\in[v]}\Big|\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})_{ms}\epsilon_{t,n}\Big|\cdot\big\|\mathbf{V}_{\mathbf{\Pi}_t^*,r}\big\|_1=O(c_Td).
\end{aligned}
\tag{S4.26}
$$

Consider now (S4.17), we have

$$
\begin{aligned}
&\Big\|\mathbf{H}_{20}-T^{-1/2}d^{a/2}\boldsymbol{\Xi}\mathbf{Y}_W\Big\|_{\max}=\max_{j,k}\Big\|\mathbf{A}_1\mathbf{A}_2\mathbf{A}_3(\mathbf{A}_{4,j,k}+\mathbf{A}_{5,j,k})-\mathbf{A}_1^0\mathbf{A}_2^0\mathbf{A}_3^0\mathbf{A}_{4,j,k}^0\Big\|_{\max}\\
&\leq\max_{j,k}\|\mathbf{A}_1\|_{\max}\|\mathbf{A}_2\|_1\|\mathbf{A}_3\|_1\|\mathbf{A}_{5,j,k}\|_1\\
&\quad+\max_{j,k}\Big\{\|\mathbf{A}_1\|_{\max}\|\mathbf{A}_2\mathbf{A}_3\mathbf{A}_{4,j,k}-\mathbf{A}_2^0\mathbf{A}_3^0\mathbf{A}_{4,j,k}^0\|_1+\|\mathbf{A}_1-\mathbf{A}_1^0\|_{\max}\|\mathbf{A}_2^0\mathbf{A}_3^0\mathbf{A}_{4,j,k}^0\|_1\Big\},\text{with}\\
&\max_{j,k}\|\mathbf{A}_2\mathbf{A}_3\mathbf{A}_{4,j,k}-\mathbf{A}_2^0\mathbf{A}_3^0\mathbf{A}_{4,j,k}^0\|_1\leq\max_{j,k}\Big\{\|\mathbf{A}_2\|_1\|\mathbf{A}_3-\mathbf{A}_3^0\|_1\|\mathbf{A}_{4,j,k}\|_1\\
&\qquad\qquad+\|\mathbf{A}_2\|_1\|\mathbf{A}_3^0\|_1\|\mathbf{A}_{4,j,k}-\mathbf{A}_{4,j,k}^0\|_1+\|\mathbf{A}_2-\mathbf{A}_2^0\|_1\|\mathbf{A}_3^0\|_1\|\mathbf{A}_{4,j,k}^0\|_1\Big\}.
\end{aligned}
$$

Together with all the rates from (S4.19) to (S4.26), we have (S4.17) true on $\mathcal{M}$.

For (S4.18), consider for any $j\in[p],k\in[l_j]\cup\{0\}$, we have on $\mathcal{M}$ that

$$
\begin{aligned}
&\Big\|\Big[\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbf{U}_{\boldsymbol{\epsilon},j,k}\boldsymbol{\Pi}_t^{*\top}\Big]\mathrm{vec}\big(\mathbf{W}_j^\top\big)\Big\|_{\max}=\max_{i\in[d]}\|(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbf{U}_{\boldsymbol{\epsilon},j,k}\boldsymbol{\Pi}_t^{*\top}\mathbf{W}_{j,i\cdot}\|_{\max}\\
&\leq\max_{i\in[d]}\|(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\mathbf{U}_{\boldsymbol{\epsilon},j,k}\|_{\max}\|\boldsymbol{\Pi}_t^*\|_\infty\|\mathbf{W}_{j,i\cdot}\|_1\\
&=O(1)\cdot\|\boldsymbol{\gamma}\|_1\cdot\max_{m,n\in[d]}\max_{s\in[v]}\Big|\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})_{ms}\epsilon_{t,n}\Big|=O(c_T),
\end{aligned}
\tag{S4.27}
$$

where the second last equality used Assumption (R1) and the result below (S4.8), and the last is similar to (S4.10). In a similar way on $\mathcal{M}$,

$$
\begin{aligned}
&\Big\|\Big[\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\big(\mathbf{U}_{\boldsymbol{\mu},j,k}-\mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},j,k})\big)\boldsymbol{\Pi}_t^{*\top}\Big]\mathrm{vec}\big(\mathbf{W}_j^\top\big)\Big\|_{\max}\\
&\leq\max_{i\in[d]}\|(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\big(\mathbf{U}_{\boldsymbol{\mu},j,k}-\mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},j,k})\big)\|_{\max}\|\boldsymbol{\Pi}_t^*\|_\infty\|\mathbf{W}_{j,i\cdot}\|_1\\
&=O\Big(\|\boldsymbol{\gamma}\|_1\|\boldsymbol{\mu}^*\|_{\max}\Big)\cdot\Big\|\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})^\top-\mathbb{E}\Big\{\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})^\top\Big\}\Big\|_{\max}=O(c_T),
\end{aligned}
\tag{S4.28}
$$

with the last line similar to (S4.25) which is also involved in the last line of the following,

$$
\begin{aligned}
&\Big\|\Big[\mathbf{I}_d\otimes(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\big(\mathbf{U}_{\mathbf{x},j,k}-\mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})\big)(\boldsymbol{\beta}^*\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}\Big]\mathrm{vec}\big(\mathbf{W}_j^\top\big)\Big\|_{\max}\\
&\leq\max_{i\in[d]}\|(\boldsymbol{\gamma}^\top\otimes\mathbf{I}_d)\big(\mathbf{U}_{\mathbf{x},j,k}-\mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})\big)\|_{\max}\|\boldsymbol{\beta}^*\otimes\mathbf{I}_d\|_1\|\boldsymbol{\Pi}_t^*\|_\infty\|\mathbf{W}_{j,i\cdot}\|_1\\
&=O\Big(\|\boldsymbol{\gamma}\|_1\Big)\max_{m,n\in[d]}\max_{s\in[v]}\max_{q\in[r]}\Big|\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})_{ms}X_{t,nq}-\mathbb{E}\Big(\frac{1}{T}\sum_{t=1}^T z_{j,k,t}(\mathbf{B}_t-\bar{\mathbf{B}})_{ms}X_{t,nq}\Big)\Big|\\
&=O(c_T).
\end{aligned}
\tag{S4.29}
$$

Combining (S4.27), (S4.28) and (S4.29), we have (S4.18) true by the following decomposition,

$$\begin{aligned}
&\left\|\mathbf{H}_{10}-T^{-1/2}d^{a/2}\mathbf{B}^{\top}\mathbf{V}\right\|_{\max}\\
\le&\max_{j,k}\left\|\left[\mathbf{I}_d\otimes(\boldsymbol{\gamma}^{\top}\otimes\mathbf{I}_d)\mathbf{U}_{\boldsymbol{\epsilon},j,k}\boldsymbol{\Pi}_t^{*\top}\right]\mathrm{vec}\big(\mathbf{W}_j^{\top}\big)\right\|_{\max}\\
&+\max_{j,k}\left\|\left[\mathbf{I}_d\otimes(\boldsymbol{\gamma}^{\top}\otimes\mathbf{I}_d)\big(\mathbf{U}_{\boldsymbol{\mu},j,k}-\mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},j,k})\big)\boldsymbol{\Pi}_t^{*\top}\right]\mathrm{vec}\big(\mathbf{W}_j^{\top}\big)\right\|_{\max}\\
&+\max_{j,k}\left\|\left[\mathbf{I}_d\otimes(\boldsymbol{\gamma}^{\top}\otimes\mathbf{I}_d)\big(\mathbf{U}_{\mathbf{x},j,k}-\mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})\big)(\boldsymbol{\beta}^{*}\otimes\mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}\right]\mathrm{vec}\big(\mathbf{W}_j^{\top}\big)\right\|_{\max}.
\end{aligned}$$

Next for F_2 and F_3, we consider first on $\mathcal{M}$,

$$\begin{aligned}
&\left\|\left\{\frac{1}{T}\sum_{t=1}^{T}\Big(\mathbf{I}_d\otimes(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\Big)\mathbf{X}_t\right\}(\mathbf{X}^{\top}\mathbf{B}^{\nu}(\mathbf{B}^{\nu})^{\top}\mathbf{X})^{-1}\mathbf{X}^{\top}\mathbf{B}^{\nu}(\mathbf{B}^{\nu})^{\top}\boldsymbol{\epsilon}^{\nu}\right\|_1\\
=&\left\|\mathbf{A}_1\mathbf{A}_2\mathbf{A}_3\frac{1}{T}\sum_{t=1}^{T}(\mathbf{B}_t-\bar{\mathbf{B}})^{\top}\boldsymbol{\epsilon}_t\right\|_1\le\left\|\mathbf{A}_1\right\|_1\left\|\mathbf{A}_2\right\|_1\left\|\mathbf{A}_3\right\|_1\cdot v\max_{j\in[r]}\left|\frac{1}{T}\sum_{t=1}^{T}\sum_{q=1}^{d}B_{t,qj}\epsilon_{t,q}\right|\\
=&O\big(d^{1+a}\cdot d^{-2}\cdot d\cdot c_Td^{\frac{1}{2}+\frac{1}{2w}}\big)=O\big(c_Td^{\frac{1}{2}+\frac{1}{2w}+a}\big),
\end{aligned}\tag{S4.30}$$

where the first equality used the fact that $\mathbf{X}_t=\mathbf{X}_t\otimes 1$, the second last used (S4.20), (S4.21), (S4.22) and $\mathcal{A}_3$ in Lemma S2. Then for F_2, we have on $\mathcal{M}$ that

$$\begin{aligned}
\|F_2\|_1\le&\frac{L^{1/2}\cdot L}{Cd^{1+a}}\Big(\left\|(\mathbf{H}_{20}-T^{-1/2}d^{a/2}\boldsymbol{\Xi}\mathbf{Y}_W)\right\|_{\max}+\left\|(T^{-1/2}d^{a/2}\mathbf{B}^{\top}\mathbf{V}-\mathbf{H}_{10})\right\|_{\max}\Big)\\
&\cdot\left\|\left\{\frac{1}{T}\sum_{t=1}^{T}\Big(\mathbf{I}_d\otimes(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\Big)\mathbf{X}_t\right\}(\mathbf{X}^{\top}\mathbf{B}^{\nu}(\mathbf{B}^{\nu})^{\top}\mathbf{X})^{-1}\mathbf{X}^{\top}\mathbf{B}^{\nu}(\mathbf{B}^{\nu})^{\top}\boldsymbol{\epsilon}^{\nu}\right\|_1\\
=&O\big(L^{3/2}\cdot d^{-1-a}\cdot c_T\cdot c_Td^{\frac{1}{2}+\frac{1}{2w}+a}\big)=O\big(c_T^2L^{3/2}d^{-\frac{1}{2}+\frac{1}{2w}}\big),
\end{aligned}\tag{S4.31}$$

where the last line used (S4.15), (S4.17), (S4.18) and (S4.30). Similarly, we have $\|F_3\|_1=O\big(c_TL^{3/2}d^{-\frac{1}{2}+\frac{1}{2w}}\big)$ on $\mathcal{M}$, and hence together with $L=O(1)$, (S4.16) and (S4.31), it holds on $\mathcal{M}$ that

$$\left\|D_2\right\|_1\le\left\|F_1\right\|_1+\left\|F_2\right\|_1+\left\|F_3\right\|_1=O\big(c_Td^{-\frac{1}{2}+\frac{1}{2w}}\big).\tag{S4.32}$$

Similar to the way we decompose D_2, we can rewrite $D_1=F_4+F_5-F_6$ where

$$\begin{aligned}
F_4=&[(\mathbf{H}_{20}-\mathbf{H}_{10})^{\top}(\mathbf{H}_{20}-\mathbf{H}_{10})]^{-1}\Big\{(\mathbf{H}_{20}-\mathbf{H}_{10})^{\top}(\mathbf{H}_{20}-\mathbf{H}_{10})\\
&-T^{-1}d^{a}(\mathbf{B}^{\top}\mathbf{V}-\boldsymbol{\Xi}\mathbf{Y}_W)^{\top}(\mathbf{B}^{\top}\mathbf{V}-\boldsymbol{\Xi}\mathbf{Y}_W)\Big\}D_1,
\end{aligned}$$

$$F_5 = [(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})]^{-1}\Big[(T^{-1/2}d^{a/2}\mathbf{B}^\top\mathbf{V}-\mathbf{H}_{10})-(T^{-1/2}d^{a/2}\boldsymbol{\Xi}\mathbf{Y}_W-\mathbf{H}_{20})\Big]^\top$$
$$\cdot T^{-1/2}d^{a/2}\mathbf{B}^\top\boldsymbol{\epsilon},$$
$$F_6 = [(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})]^{-1}(\mathbf{H}_{20}-\mathbf{H}_{10})^\top\cdot T^{-1/2}d^{a/2}\mathbf{B}^\top\boldsymbol{\epsilon}.$$

From (S4.16), it is direct that $F_4 = O\big(c_T L^{3/2}d^{1-a}\|D_1\|_1\big)$ on $\mathcal{M}$. Moreover, (S4.17) and (S4.18) imply that F_5 has a smaller rate than that of F_6. Given $L=O(1)$, we next construct the asymptotic normality of $\boldsymbol{\alpha}^\top F_6$ for any given non-zero $\boldsymbol{\alpha}\in\mathbb{R}^L$ with $\|\boldsymbol{\alpha}\|_1\le c<\infty$.

Denote by $\mathbf{R}_1 := [(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})]^{-1}(\mathbf{H}_{20}-\mathbf{H}_{10})^\top$, we have

$$\begin{aligned}
\boldsymbol{\alpha}^\top F_6 &= \frac{1}{T}\boldsymbol{\alpha}^\top\mathbf{R}_1\Big(\mathbf{I}_d\otimes\big\{(\mathbf{B}_1-\bar{\mathbf{B}},\dots,\mathbf{B}_T-\bar{\mathbf{B}})(\mathbf{I}_T\otimes\boldsymbol{\gamma})\big\}\Big)\mathrm{vec}\big((\boldsymbol{\epsilon}_1,\dots,\boldsymbol{\epsilon}_T)^\top\big)\\
&= \frac{1}{T}\boldsymbol{\alpha}^\top\mathbf{R}_1\Big(\mathbf{I}_d\otimes\big\{(\mathbf{B}_1-\bar{\mathbf{B}})\boldsymbol{\gamma},\dots,(\mathbf{B}_T-\bar{\mathbf{B}})\boldsymbol{\gamma}\big\}\Big)\sum_{t=1}^T\Big\{\boldsymbol{\epsilon}_t\otimes\big(\mathbb{1}_{\{j=t\}}\big)_{j\in[T]}\Big\}\\
&= \frac{1}{T}\sum_{t=1}^T\boldsymbol{\alpha}^\top\mathbf{R}_1\Big(\boldsymbol{\epsilon}_t\otimes(\mathbf{B}_t-\bar{\mathbf{B}})\boldsymbol{\gamma}\Big) = \frac{1}{T}\sum_{t=1}^T\boldsymbol{\alpha}^\top\mathbf{R}_1\Big(\boldsymbol{\epsilon}_t\otimes[\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma}(1+o_P(1))\Big),
\end{aligned}$$

where the vector $\big(\mathbb{1}_{\{j=t\}}\big)_{j\in[T]}\in\mathbb{R}^T$ has value $\mathbb{1}_{\{j=t\}}$ at each j-th entry, and the last equality used $\mathcal{A}_4$ in Lemma S2. Hence to apply Theorem 3 (ii) of Wu (2011), we need to show

$$\sum_{t\ge 0}\Big\|P_0\Big\{\boldsymbol{\alpha}^\top\mathbf{R}_1\big(\boldsymbol{\epsilon}_t\otimes[\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma}\big)\Big\}\Big\|_2<\infty, \tag{S4.33}$$

where $P_0(\cdot) := \mathbb{E}_0(\cdot)-\mathbb{E}_{-1}(\cdot)$ and $\mathbb{E}_i(\cdot) := \mathbb{E}(\cdot\,|\,\sigma(\mathcal{G}_i,\mathcal{H}_i))$. Notice that

$$\begin{aligned}
&\Big\|P_0\Big\{\boldsymbol{\alpha}^\top\mathbf{R}_1\big(\boldsymbol{\epsilon}_t\otimes[\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma}\big)\Big\}\Big\|_2\\
&= \Big\|\boldsymbol{\alpha}^\top\mathbf{R}_1\Big\{P_0(\boldsymbol{\epsilon}_t)\otimes\mathbb{E}_0([\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma})\Big\}+\boldsymbol{\alpha}^\top\mathbf{R}_1\Big\{\mathbb{E}_{-1}(\boldsymbol{\epsilon}_t)\otimes P_0([\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma})\Big\}\Big\|_2\\
&\le \Big\{2\boldsymbol{\alpha}^\top\mathbf{R}_1\Big\{\mathbb{E}\big(P_0(\boldsymbol{\epsilon}_t)P_0(\boldsymbol{\epsilon}_t)^\top\big)\otimes\mathbb{E}\big(\mathbb{E}_0([\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma})\mathbb{E}_0(\boldsymbol{\gamma}^\top[\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t)]^\top)\big)\Big\}\mathbf{R}_1^\top\boldsymbol{\alpha}\Big\}^{\frac{1}{2}}\\
&+\Big(2\boldsymbol{\alpha}^\top\mathbf{R}_1\Big\{\mathbb{E}\big(\mathbb{E}_{-1}(\boldsymbol{\epsilon}_t)\mathbb{E}_{-1}(\boldsymbol{\epsilon}_t)^\top\big)\otimes\mathbb{E}\big(P_0([\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma})P_0(\boldsymbol{\gamma}^\top[\mathbf{B}_t-\mathbb{E}(\mathbf{B}_t)]^\top)\big)\Big\}\mathbf{R}_1^\top\boldsymbol{\alpha}\Big)^{\frac{1}{2}}\\
&= O\Big(\|\boldsymbol{\alpha}\|_1\|\mathbf{R}_1\|_\infty\Big)\cdot\Big(\max_{j\in[d]}\|P_0(\epsilon_{t,j})\|_2\cdot\max_{j\in[d]}\mathrm{Var}^{1/2}(\mathbf{B}_{t,j\cdot}^\top\boldsymbol{\gamma})+\sigma_{\max}\max_{j\in[d]}\|P_0(\mathbf{B}_{t,j\cdot}^\top\boldsymbol{\gamma})\|_2\Big)\\
&= O\Big(\max_{j\in[d]}\|P_0^\epsilon(\epsilon_{t,j})\|_2+\max_{j\in[d]}\max_{k\in[v]}\|P_0^b(B_{t,jk})\|_2\Big),
\end{aligned}$$

where the second last equality used $\mathrm{Var}(\cdot)=\mathrm{Var}(\mathbb{E}_i(\cdot))+\mathbb{E}(\mathrm{Var}_i(\cdot))\ge\mathrm{Var}(\mathbb{E}_i(\cdot))$, and the last used Assumption (R2) and $\|\mathbf{R}_1\|_\infty=O(1)$ which is implied from (S4.15), $\|\mathbf{H}_{10}\|_1=O(d)$ and

$\left\|\mathbf{H}_{20}\right\|_1 = O(d^{1+a})$. With Assumption (R7), (S4.33) is true. Therefore, with definition

$$s_1 := \boldsymbol{\alpha}^\top \mathbf{R}_1 \boldsymbol{\Sigma} \mathbf{R}_1^\top \boldsymbol{\alpha}, \boldsymbol{\Sigma} := \sum_\tau \mathbb{E}(\boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_{t+\tau}^\top) \otimes \mathbb{E}\big[(\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))\boldsymbol{\gamma}\boldsymbol{\gamma}^\top (\mathbf{B}_{t+\tau} - \mathbb{E}(\mathbf{B}_t))^\top\big], \tag{S4.34}$$

we have by Theorem 3 (ii) of Wu (2011) that

$$T^{1/2} s_1^{-1/2} \boldsymbol{\alpha}^\top F_6 \xrightarrow{D} \mathcal{N}(0,1).$$

Then equivalently we have

$$T^{1/2} (\mathbf{R}_1 \boldsymbol{\Sigma} \mathbf{R}_1^\top)^{-1/2} F_6 \xrightarrow{D} \mathcal{N}(\mathbf{0}, \mathbf{I}_L), \tag{S4.35}$$

so that F_6 is at least $T^{1/2} d^{(1+a-b)/2}$-convergent which used $\lambda_{\max}(\mathbf{R}_1 \mathbf{R}_1^\top) = O(d^{-1-a})$ from (S4.15), and all eigenvalues of $d^{-b}\boldsymbol{\Sigma}$ uniformly bounded from 0 and infinity by Assumption (R8). Hence, $\left\|D_1\right\|_1 = O\big(\|F_6\|_1\big) = O\big(T^{-1/2} d^{-(1+a-b)/2}\big)$ on $\mathcal{M}$, and by (S4.32) we have

$$\left\|\widetilde{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\right\|_1 = O_P\Big(\left\|D_1\right\|_1 + \left\|D_2\right\|_1\Big) = O_P\big(T^{-1/2} d^{-(1+a-b)/2} + c_T d^{-\frac{1}{2} + \frac{1}{2w}}\big) = O_P\big(c_T d^{-\frac{1}{2} + \frac{1}{2w}}\big),$$

where the last equality used Assumption (R10). With the above plugged into (S4.12), the proof of Theorem 1 is complete. $\square$

Proof of Theorem 2. By the KKT condition, $\widehat{\boldsymbol{\phi}}$ is a solution to the adaptive LASSO problem in (2.15) if and only if there exists a subgradient

$$\mathbf{h} = \partial(\mathbf{u}^\top |\widehat{\boldsymbol{\phi}}|) = \left\{ \mathbf{h} \in \mathbb{R}^L : \begin{cases} h_i = u_i \operatorname{sign}(\widehat{\phi}_i), & \widehat{\phi}_i \neq 0; \\ |h_i| \leq u_i, & \text{otherwise.} \end{cases} \right\},$$

such that differentiating the expression in (2.15) with respect to $\boldsymbol{\phi}$, we have

$$T^{-1}(\boldsymbol{\Xi}\mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\boldsymbol{\Xi}\mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})\boldsymbol{\phi} + T^{-1}(\boldsymbol{\Xi}\mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\mathbf{B}^\top \mathbf{y} - \boldsymbol{\Xi}\mathbf{y}^\nu) = -\lambda \mathbf{h}.$$

Substituting (2.11) in the above, we arrive at

$$\begin{aligned} -\lambda \mathbf{h} &= T^{-1}(\boldsymbol{\Xi}\mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\boldsymbol{\Xi}\mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})\boldsymbol{\phi} \\ &\quad + T^{-1}(\boldsymbol{\Xi}\mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \big(\mathbf{B}^\top \mathbf{V}\boldsymbol{\phi}^* + \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}^*} \operatorname{vec}\big(\mathbf{I}_d\big) + \mathbf{B}^\top \boldsymbol{\epsilon} - \boldsymbol{\Xi}\mathbf{y}^\nu\big) \end{aligned}$$

$$
\begin{aligned}
&= T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \mathbf{V}(\boldsymbol{\phi}^* - \boldsymbol{\phi}) + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \boldsymbol{\epsilon} \\
&\quad + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}^*} \mathrm{vec}\big(\mathbf{I}_d\big) \\
&\quad + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\Xi \mathbf{Y}_W \boldsymbol{\phi}^* - \Xi \mathbf{y}^\nu + \Xi \mathbf{Y}_W(\boldsymbol{\phi} - \boldsymbol{\phi}^*)) \\
&= T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})(\boldsymbol{\phi} - \boldsymbol{\phi}^*) + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \boldsymbol{\epsilon} \\
&\quad + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}^* - \boldsymbol{\beta}(\boldsymbol{\phi}^*)} \mathrm{vec}\big(\mathbf{I}_d\big),
\end{aligned}
$$

where the last equality used the fact that $\mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}(\boldsymbol{\phi}^*)} \mathrm{vec}\big(\mathbf{I}_d\big) = \Xi \mathbf{y}^\nu - \Xi \mathbf{Y}_W \boldsymbol{\phi}^*$ from (S1.3). Then we may conclude that there exists a sign-consistent solution $\widehat{\boldsymbol{\phi}}$ if and only if

$$
\begin{cases}
-\lambda \mathbf{h}_H = T^{-1}(\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H)^\top (\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H)(\widehat{\boldsymbol{\phi}} - \boldsymbol{\phi}^*) \\
\qquad + T^{-1}(\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H)^\top \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}^* - \boldsymbol{\beta}(\boldsymbol{\phi}^*)} \mathrm{vec}\big(\mathbf{I}_d\big) + T^{-1}(\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H)^\top \mathbf{B}^\top \boldsymbol{\epsilon}, \\
\lambda \mathbf{u}_{H^c} \geq \Big| T^{-1}(\Xi \mathbf{Y}_{W,H^c} - \mathbf{B}^\top \mathbf{V}_{H^c})^\top \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}^* - \boldsymbol{\beta}(\boldsymbol{\phi}^*)} \mathrm{vec}\big(\mathbf{I}_d\big) + T^{-1}(\Xi \mathbf{Y}_{W,H^c} - \mathbf{B}^\top \mathbf{V}_{H^c})^\top \mathbf{B}^\top \boldsymbol{\epsilon} \Big|,
\end{cases}
\tag{S4.36}
$$

where $\mathbf{A}_H$ and $\mathbf{a}_H$ denote the corresponding submatrix $\mathbf{A}$ with columns restricted on the set H and subvector $\mathbf{a}$ with entries restricted on the set H, respectively. Similarly $(\cdot)_{H^c}$ is defined. Consider the first equation in (S4.36), similar to how D_2 is decomposed in the proof of Theorem 1, we write $\widehat{\boldsymbol{\phi}} - \boldsymbol{\phi}^* = \sum_{j=1}^4 I_{\phi,j}$ where

$$
\begin{aligned}
I_{\phi,1} &= [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} \Big\{ (\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H \\
&\qquad - T^{-1} d^a (\mathbf{B}^\top \mathbf{V}_H - \Xi \mathbf{Y}_{W,H})^\top (\mathbf{B}^\top \mathbf{V}_H - \Xi \mathbf{Y}_{W,H}) \Big\} (\widehat{\boldsymbol{\phi}} - \boldsymbol{\phi}^*), \\
I_{\phi,2} &= -[(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} d^a \lambda \mathbf{h}_H, \\
I_{\phi,3} &= [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} T^{-1} d^a (\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H)^\top \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}(\boldsymbol{\phi}^*) - \boldsymbol{\beta}^*} \mathrm{vec}\big(\mathbf{I}_d\big), \\
I_{\phi,4} &= [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} T^{-1} d^a (\mathbf{B}^\top \mathbf{V}_H - \Xi \mathbf{Y}_{W,H})^\top \mathbf{B}^\top \boldsymbol{\epsilon}.
\end{aligned}
$$

Similar to F_1 in the proof of Theorem 1, we may derive that

$$
\big\| I_{\phi,1} \big\|_{\max} = O_P\Big(c_T d^{1-a} \big\| \widehat{\boldsymbol{\phi}} - \boldsymbol{\phi}^* \big\|_{\max} \Big) = o_P\Big(\big\| \widehat{\boldsymbol{\phi}} - \boldsymbol{\phi}^* \big\|_{\max} \Big), \tag{S4.37}
$$

where the first equality used the fact that (R1) implies for a positive constant u that $\sigma_{|H|}^2\{(\mathbf{D}_W)_H\} \geq du > 0$ uniformly as $d \to \infty$, and the conditions in the statement of Theorem 2, and the second used (R10). Similarly, with Assumption (R9) we have

$$\left\|I_{\phi,2}\right\|_{\max} = O_P\left(d^{-1-a} \cdot d^{a} \cdot \lambda\right) = O_P\left(c_T d^{-1}\right). \tag{S4.38}$$

For $I_{\phi,4}$, we may decompose it as the following with the second term dominating the first term similarly to F_5 and F_6 in the proof of Theorem 1,

$$\begin{aligned}
I_{\phi,4} = \Big(& [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^{\top} (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} \\
& \cdot \Big[(T^{-1/2} d^{a/2} \mathbf{B}^{\top} \mathbf{V}_H - \mathbf{H}_{10,H}) - (T^{-1/2} d^{a/2} \boldsymbol{\Xi} \mathbf{Y}_{W,H} - \mathbf{H}_{20,H}) \Big]^{\top} \cdot T^{-1/2} d^{a/2} \mathbf{B}^{\top} \boldsymbol{\epsilon} \Big) \\
& - \Big([(\mathbf{H}_{20} - \mathbf{H}_{10})_H^{\top} (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} (\mathbf{H}_{20} - \mathbf{H}_{10})_H^{\top} \cdot T^{-1/2} d^{a/2} \mathbf{B}^{\top} \boldsymbol{\epsilon} \Big).
\end{aligned}$$

The second term in the above has rate $T^{-1/2} d^{-(1+a-b)/2}$ by exactly the same way to construct asymptotic normality of F_6 in (S4.35), except for the restriction to the set H here (proof omitted). Thus,

$$\left\|I_{\phi,4}\right\|_{\max} = O_P\left(T^{-1/2} d^{-(1+a-b)/2}\right). \tag{S4.39}$$

We next construct the asymptotic normality of $I_{\phi,3}$ and show its rate is of order $T^{-1/2} d^{-(1-b)/2}$ which is dominating over those of $I_{\phi,1}$, $I_{\phi,2}$ and $I_{\phi,4}$ by Assumption (R10). Recall $\mathbf{R}_H = [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^{\top} (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} (\mathbf{H}_{20} - \mathbf{H}_{10})_H^{\top}$, and let non-zero $\boldsymbol{\alpha} \in \mathbb{R}^{|H|}$ such that $\|\boldsymbol{\alpha}\|_1 \leq c < \infty$. Then we have

$$\begin{aligned}
& \boldsymbol{\alpha}^{\top} I_{\phi,3} = \boldsymbol{\alpha}^{\top} \mathbf{R}_H T^{-1} (\mathbf{B}_{\boldsymbol{\gamma}} - \bar{\mathbf{B}}_{\boldsymbol{\gamma}})^{\top} \mathbf{X}_{\boldsymbol{\beta}(\boldsymbol{\phi}^*) - \boldsymbol{\beta}^*} \mathrm{vec}\big(\mathbf{I}_d\big)(1 + o_P(1)) \\
&= \boldsymbol{\alpha}^{\top} \mathbf{R}_H T^{-1} \Big(\mathbf{I}_d \otimes \big\{ (\mathbf{B}_1 - \bar{\mathbf{B}}, \ldots, \mathbf{B}_T - \bar{\mathbf{B}}) \\
&\quad \cdot (\mathbf{I}_T \otimes \boldsymbol{\gamma} (\boldsymbol{\beta}(\boldsymbol{\phi}^*) - \boldsymbol{\beta}^*)^{\top}) (\mathbf{X}_1, \ldots, \mathbf{X}_T)^{\top} \big\} \Big) \mathrm{vec}\big(\mathbf{I}_d\big)(1 + o_P(1)) \\
&= \boldsymbol{\alpha}^{\top} \mathbf{R}_H \frac{1}{T} \sum_{t=1}^{T} \mathrm{vec}\Big((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)) \boldsymbol{\gamma} (\boldsymbol{\beta}(\boldsymbol{\phi}^*) - \boldsymbol{\beta}^*)^{\top} \mathbf{X}_t^{\top} \Big)(1 + o_P(1)) \\
&= \boldsymbol{\alpha}^{\top} \mathbf{R}_H \mathrm{vec}\Big(\frac{1}{T} \sum_{t=1}^{T} \Big[\boldsymbol{\gamma}^{\top} (\mathbf{B}_{t,i\cdot} - \mathbb{E}(\mathbf{B}_{t,i\cdot})) \mathbf{X}_{t,j\cdot}^{\top} (\boldsymbol{\beta}(\boldsymbol{\phi}^*) - \boldsymbol{\beta}^*) \Big]_{i,j \in [d]} \Big)(1 + o_P(1)) \\
&= \boldsymbol{\alpha}^{\top} \mathbf{R}_H \mathbf{S}_{\boldsymbol{\gamma}} (\boldsymbol{\beta}(\boldsymbol{\phi}^*) - \boldsymbol{\beta}^*)(1 + o_P(1)),
\end{aligned}$$

where the third last equality used $\mathcal{A}_4$ in Lemma S2 and the last used $\mathcal{A}_1$. From (S4.6) and Lemma

S4, we have

$$
\begin{aligned}
&T^{1/2}(\mathbf{R}_\beta\boldsymbol{\Sigma}_\beta\mathbf{R}_\beta^\top)^{-1/2}(\boldsymbol{\beta}(\boldsymbol{\phi}^*)-\boldsymbol{\beta}^*)\\
=&T^{\frac{1}{2}}(\mathbf{R}_\beta\boldsymbol{\Sigma}_\beta\mathbf{R}_\beta^\top)^{-\frac{1}{2}}\Big\{\big(\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\mathbf{X}\big)^{-1}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu\Big\}\\
=&T^{\frac{1}{2}}(\mathbf{R}_\beta\boldsymbol{\Sigma}_\beta\mathbf{R}_\beta^\top)^{-\frac{1}{2}}\Big\{\big[\mathbb{E}(\mathbf{X}_t^\top\mathbf{B}_t)\mathbb{E}(\mathbf{B}_t^\top\mathbf{X}_t)\big]^{-1}\frac{1}{T^2}\mathbf{X}^\top\mathbf{B}^\nu(\mathbf{B}^\nu)^\top\boldsymbol{\epsilon}^\nu(1+o(1))\Big\}\xrightarrow{D}\mathcal{N}(\mathbf{0},\mathbf{I}_r).
\end{aligned}
$$

Define $s_3:=\boldsymbol{\alpha}^\top\mathbf{R}_H\mathbf{S}_{\boldsymbol{\gamma}}\mathbf{R}_\beta\boldsymbol{\Sigma}_\beta\mathbf{R}_\beta^\top\mathbf{S}_{\boldsymbol{\gamma}}^\top\mathbf{R}_H^\top\boldsymbol{\alpha}$, we hence have $T^{1/2}s_3^{-1/2}\boldsymbol{\alpha}^\top I_{\phi,3}\xrightarrow{D}\mathcal{N}(0,1)$ and equivalently

$$
T^{1/2}(\mathbf{R}_H\mathbf{S}_{\boldsymbol{\gamma}}\mathbf{R}_\beta\boldsymbol{\Sigma}_\beta\mathbf{R}_\beta^\top\mathbf{S}_{\boldsymbol{\gamma}}^\top\mathbf{R}_H^\top)^{-1/2}I_{\phi,3}\xrightarrow{D}\mathcal{N}(\mathbf{0},\mathbf{I}_{|H|}).
$$

As shown in the proof of Lemma S4, the eigenvalues of $\mathbf{R}_\beta\boldsymbol{\Sigma}_\beta\mathbf{R}_\beta^\top$ are of order d^{b-1}. Similar to $\mathbf{R}_1$ in the proof of Theorem 1, $\lambda_{\max}(\mathbf{R}_H\mathbf{R}_H^\top)=O(d^{-1-a})$. We also have $\lambda_{\max}(\mathbf{S}_{\boldsymbol{\gamma}}^\top\mathbf{S}_{\boldsymbol{\gamma}})=O(d^{1+a})$ by Assumption (R5). Combining them, we have

$$
\begin{aligned}
&\|\boldsymbol{\alpha}\|_1^2\lambda_{\min}(\mathbf{R}_H\mathbf{S}_{\boldsymbol{\gamma}}\mathbf{S}_{\boldsymbol{\gamma}}^\top\mathbf{R}_H^\top)\lambda_{\min}(\mathbf{R}_\beta\boldsymbol{\Sigma}_\beta\mathbf{R}_\beta^\top)\\
\le& s_3\le\|\boldsymbol{\alpha}\|_1^2\lambda_{\max}(\mathbf{R}_H\mathbf{R}_H^\top)\lambda_{\max}(\mathbf{S}_{\boldsymbol{\gamma}}^\top\mathbf{S}_{\boldsymbol{\gamma}})\lambda_{\max}(\mathbf{R}_\beta\boldsymbol{\Sigma}_\beta\mathbf{R}_\beta^\top),
\end{aligned}
$$

with the right hand side of order d^{b-1}. The left hand side is of the same order by the assumption in the statement of Theorem 2 that $\mathbf{R}_H\mathbf{S}_{\boldsymbol{\gamma}}\mathbf{S}_{\boldsymbol{\gamma}}^\top\mathbf{R}_H^\top$ has the smallest eigenvalue of constant order. Thus, s_3 is of order exactly d^{b-1} and hence $\boldsymbol{\alpha}^\top I_{\phi,3}$ has order exactly $T^{-1/2}d^{-(1-b)/2}$. It implies $I_{\phi,3}$ is the leading term in $\widehat{\boldsymbol{\phi}}-\boldsymbol{\phi}^*$ whose asymptotic normality therefore holds.

As $I_{\phi,1}$ to $I_{\phi,4}$ are all $o_P(1)$, we conclude $\text{sign}(\widehat{\boldsymbol{\phi}}_H)=\text{sign}(\boldsymbol{\phi}_H^*)$. It remains to show the second part in (S4.36) for the zero consistency of $\widehat{\boldsymbol{\phi}}_{H^c}$.

To this end, notice similar to $I_{\phi,3}$ but with restriction on the set H^c, we have

$$
\begin{aligned}
&\big\|T^{-1}(\boldsymbol{\Xi}\mathbf{Y}_{W,H^c}-\mathbf{B}^\top\mathbf{V}_{H^c})^\top\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}(\boldsymbol{\phi}^*)}\text{vec}\big(\mathbf{I}_d\big)\big\|_{\max}\\
=&O_P\big(T^{-1/2}d^{-(1-b)/2}\cdot d^{-a}\cdot d^{1+2a}\big)=O_P\big(T^{-1/2}d^{\frac{1}{2}+\frac{b}{2}+a}\big),
\end{aligned}
$$

which used $\big\|(\mathbf{H}_{20}-\mathbf{H}_{10})^\top(\mathbf{H}_{20}-\mathbf{H}_{10})\big\|_{\max}\le\sigma_1^2(\mathbf{H}_{20}-\mathbf{H}_{10})=O(d^{1+2a})$ similarly to the steps above (S4.15). In the same manner, we also have from $I_{\phi,4}$ that

$$
\big\|T^{-1}(\boldsymbol{\Xi}\mathbf{Y}_{W,H^c}-\mathbf{B}^\top\mathbf{V}_{H^c})^\top\mathbf{B}^\top\boldsymbol{\epsilon}\big\|_{\max}=O_P\big(T^{-1/2}d^{\frac{1}{2}+\frac{b}{2}+\frac{a}{2}}\big).
$$

The left hand side of the second inequality in (S4.36) has minimum value of

$$\frac{\lambda}{\|\widetilde{\boldsymbol{\phi}}_{H^c}\|_{\max}} \geq \frac{\lambda}{\|\widetilde{\boldsymbol{\phi}}_{H^c} - \boldsymbol{\phi}^*_{H^c}\|_{\max}} \geq \frac{\lambda}{\|\widetilde{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_{\max}},$$

so it suffices to show

$$\big(T^{-1/2} d^{\frac{1}{2}+\frac{b}{2}+a}\big) \cdot \|\widetilde{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_{\max} = o_P(c_T),$$

which is true by Assumption (R10) and Theorem 1 in which each entry of F_6 can be shown to be asymptotically normal. This completes the proof of Theorem 2. □

Proof of Theorem 3. We have

$$\begin{aligned} \|\widehat{\mathbf{W}}_t - \mathbf{W}^*_t\|_\infty &= \Big\| \sum_{j=1}^{p} \Big[(\widehat{\phi}_{j,0} - \phi^*_{j,0}) + \sum_{k=1}^{l_j} (\widehat{\phi}_{j,k} - \phi^*_{j,k}) z_{j,k,t} \Big] \mathbf{W}_j \Big\|_\infty \\ &= O_P\Big(\|\widehat{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_1 \cdot \max_j \|\mathbf{W}_j\|_\infty \Big) = O_P\big(T^{-1/2} d^{-(1-b)/2}\big), \end{aligned} \tag{S4.40}$$

where the last equality used Theorem 2, Assumptions (M2) (or (M2')) and (R1). Observe that we have similarly $\|\widehat{\mathbf{W}}_t - \mathbf{W}^*_t\|_1 = O_P\big(T^{-1/2} d^{-(1-b)/2}\big)$ by Assumption (R1), and hence

$$\|\widehat{\mathbf{W}}_t - \mathbf{W}^*_t\| \leq \Big(\|\widehat{\mathbf{W}}_t - \mathbf{W}^*_t\|_1 \|\widehat{\mathbf{W}}_t - \mathbf{W}^*_t\|_\infty \Big)^{1/2} = O_P\big(T^{-1/2} d^{-(1-b)/2}\big).$$

With $\boldsymbol{\Pi}^*_t$ defined in (S4.8), we can decompose

$$\begin{aligned} \widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^* &= \frac{1}{T} \sum_{t=1}^{T} \Big\{ (\mathbf{I}_d - \boldsymbol{\Lambda}_t \widehat{\boldsymbol{\Phi}}) \mathbf{y}_t - \mathbf{X}_t \widehat{\boldsymbol{\beta}} \Big\} - \frac{1}{T} \sum_{t=1}^{T} \Big\{ (\mathbf{I}_d - \boldsymbol{\Lambda}_t \boldsymbol{\Phi}^*) \mathbf{y}_t - \mathbf{X}_t \boldsymbol{\beta}^* - \boldsymbol{\epsilon}_t \Big\} \\ &= \frac{1}{T} \sum_{t=1}^{T} \Big\{ \big(\mathbf{W}^*_t - \widehat{\mathbf{W}}_t\big) \big(\boldsymbol{\Pi}^*_t \boldsymbol{\mu}^* + \boldsymbol{\Pi}^*_t \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\Pi}^*_t \boldsymbol{\epsilon}_t \big) \Big\} + \bar{\mathbf{X}}(\boldsymbol{\beta}^* - \widehat{\boldsymbol{\beta}}) + \bar{\boldsymbol{\epsilon}}, \end{aligned}$$

so that combining (S4.40), Lemma S2 and Theorem 1, we have

$$\begin{aligned} \|\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\|_{\max} = O_P\Big\{ \max_t \|\widehat{\mathbf{W}}_t - \mathbf{W}^*_t\|_\infty \Big(\|\boldsymbol{\mu}^*\|_{\max} + c_T \|\boldsymbol{\beta}^*\|_{\max} + c_T \Big) \\ + c_T \|\boldsymbol{\beta}^* - \widehat{\boldsymbol{\beta}}\|_1 + c_T \Big\} = O_P(c_T). \end{aligned}$$

This completes the proof of Theorem 3. □

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